STPM Physics · Revision notes

PhysiQ

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SEMESTER 2 NOTES

Electricity & Magnetism · Chapters 13–19

7 chapters · 34 topics. Definitions and laws, formula tables with symbols and units, tips to answer, and the Section A traps and tricks that cost marks in the multiple-choice paper.

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Trap: a wrong option that looks right — the slip examiners rely on.

Trick: a quick check or shortcut for the multiple-choice paper.

Section A · Semester 2

SECTION A GUIDE

BEFORE YOU START: THE 6-STEP SECTION A ROUTINE

- **Read the LAST line first** — know what is being asked (a value, a ratio, a direction, "which is NOT true") before reading the data.
- **Circle the trigger words**: NOT, only, always, never, increases, decreases, ratio, "per unit", peak / r.m.s., radius / diameter.
- **Convert units immediately**: $mm^2 \rightarrow \times 10^{-6}\ m^2$, $cm \rightarrow \times 10^{-2}\ m$, $\mu = 10^{-6}$, $n = 10^{-9}$, $p = 10^{-12}$, °C → K, minutes → seconds.
- **Estimate first**: do a rough order-of-magnitude calculation (powers of ten) — usually two options are wildly off and can be removed before you do any precise arithmetic.
- **Match the answer to the distractors**: examiners build wrong options from a forgotten square, a missing factor of 2, an inverted ratio, a wrong sign, or an un-converted unit. If your answer matches a "classic slip" value, re-check that step.
- **Time-box**: about 1½ minutes per question. Mark a hard one, move on, return later. Never leave a question blank — eliminate what you can and choose.

UNIVERSAL TRICKS THAT WORK IN EVERY SEMESTER

Trick: Ratio method — for "if X is doubled/tripled, what happens to Y?", write $Y \propto$ (formula), then substitute the factors (e.g. $\frac{2}{3^2} = \frac{2}{9}$). Never calculate with real numbers when a ratio will do.

Trick: Dimension / unit check — if the unit of your final expression is not the unit asked for, it is wrong. Quick way to remove 1–2 options.

Trick: Extreme-case check — imagine $r \rightarrow \infty$, $t \rightarrow 0$ or $t \rightarrow \infty$. Does the option behave sensibly (e.g. a discharging capacitor must end at zero)?

Trick: Direction/sign questions — draw a 2-second sketch with axes. Do not trust your head for left-hand/right-hand rules or signs.

Trick: "Which statement is correct" questions — test each option separately and write ✓ / × next to it; one clearly false word (e.g. "always", "only") normally makes an option false.

Trick: Graph questions — decide first what the gradient and the area under the graph mean physically, then read the options.

SEMESTER 2 — HOTSPOTS (ELECTRICITY, MAGNETISM, A.C.)

Trap: Peak vs r.m.s.: $V_{rms} = \frac{V_0}{\sqrt{2}}$. Mean power uses r.m.s. values: $P = \frac{V_{rms}^2}{R}$. If the question gives $V = V_0 \sin \omega t$, read V_0 first, then divide by $\sqrt{2}$.

Trap: ω is not f: $f = \frac{\omega}{2\pi}$. Reactances use 2πf ($X_L = \omega L$, $X_C = \frac{1}{\omega C}$).

Trap: Series/parallel is REVERSED for capacitors: parallel adds ($C = C_1 + C_2$), series uses $\frac{1}{C} = \frac{\Sigma 1}{C}$. Resistors are the opposite.

Trap: Constant V or constant Q? A capacitor connected to a battery keeps V; an isolated charged capacitor keeps Q. Ask this before using $U = \frac{1}{2}CV^2$ or $\frac{Q^2}{2C}$.

Trap: Radius vs diameter in $A = \pi r^2$ (drift velocity, resistance, current density). Convert mm^2 correctly (10^{-6} m^2).

Trap: Stretching a wire: volume is constant, so length × n means area ÷ n and R × n^2.

Trap: Terminal p.d. ≠ e.m.f.: $V = E - Ir$ whenever a current flows. They are equal only for $I = 0$.

Trap: Magnetic flux uses the angle to the NORMAL: $\Phi = BA\cos\theta$. Force on a conductor uses the angle to the FIELD: $F = BIL\sin\theta$.

Trap: Solenoid: $B = \mu_0 nI$ uses turns per metre, not total turns.

Trap: R–L and R–C circuits: V_R, V_L, V_C add as phasors (90° apart), never arithmetically; $Z = \sqrt{R^2 + X^2}$.

Trap: Lenz's law: the induced effect OPPOSES the change in flux — not the flux itself.

Trick: RC circuits: after 1τ a discharging capacitor is at 37%; a charging one at 63%. After 2τ: 14% / 86%. Time to reach a fraction f of charge: $t = -\tau \ln(1 - f)$.

Trick: Currents in same direction attract; opposite directions repel (parallel wires) — and the two forces are always equal in size.

Trick: Fleming's left-hand rule for force (motor): First finger field, seCond finger Current, thuMb Motion. Right-hand rule is for generators / induced current.

Trick: $\frac{e}{m}$: $eV = \frac{1}{2}mv^2$ and $evB = \frac{mv^2}{r}$ combine to $\frac{e}{m} = 2V/(B^2r^2)$.

Trick: $X_L \propto f$, $X_C \propto \frac{1}{f}$: a capacitor passes high frequencies and blocks d.c.; an inductor passes d.c. and resists high frequencies.

> **Trick: Phase memory**: "ELI the ICE man" — in an inductor voltage (E) leads current (I); in a capacitor current leads voltage.

Chapter 13

ELECTROSTATICS

13.1 Coulomb's Law

DEFINITIONS & LAWS

- **Coulomb's law** states that the electrostatic force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.
- The force acts along the line joining the two charges; it is repulsive for like charges and attractive for unlike charges. The two forces form a Newton's third law pair (equal in magnitude, opposite in direction).
- **Principle of superposition**: the net force on a charge is the vector sum of the forces exerted on it by each of the other charges separately.
- **Relative permittivity** ε_r of a medium is the ratio of the force between two charges in a vacuum to the force between the same charges, at the same separation, in the medium ($\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}$).
- **Charge is quantised**: Q = ne, where n is an integer.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Coulomb's law (vacuum)	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2} = \frac{kQ_1Q_2}{r^2}$	F = electrostatic force (N); Q_1, Q_2 = point charges (C); r = separation (m); $k = \frac{1}{4\pi\varepsilon_0} = 8.99\times10^9$ N m^2 C^{-2}; $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
Coulomb's law in a medium	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0\varepsilon_r r^2}$	ε_r = relative permittivity of the medium (no unit); other symbols as above
Force in an electric field	$F = qE$	F = force (N); q = charge (C); E = electric field strength (N C^{-1} or V m^{-1})
Acceleration of a charge	$a = F/m$	a = acceleration (m s^{-2}); F = force (N); m = mass (kg)
Charged spheres on threads	$\tan\theta = \frac{F}{mg}$	Two small identical charged spheres on threads; θ = angle of thread to the vertical; F = horizontal electric force (N), balanced by the tension and the weight mg; m = mass (kg); g = acceleration of free fall (m s^{-2})

- e = 1.60×10^{-19} C; 1 μC = 10^{-6} C; 1 nC = 10^{-9} C.

TIPS TO ANSWER

- State the law or formula first, then substitute with all quantities in SI units (convert μC, nC, cm, mm), then give the answer with unit to 3 s.f.
- State the nature of the force: "repulsive" if the charges have the same sign, "attractive" if the signs are opposite.
- For several charges, draw the force on the chosen charge from each of the others, add them as vectors (along a line: take one direction as positive) and state the direction of the resultant.
- For a charged-sphere or thread problem, draw the free-body diagram (T, mg, F), resolve and write $\tan\theta = \frac{F}{mg}$.
- For a null point, equate the magnitudes of the two forces (or fields) and solve for the distance.
- Use ε_0 from the list of constants $\frac{1}{4\pi\varepsilon_0}$, not a rounded $k = 9 \times 10^9$ N m^2 C^{-2}. A slightly different final answer from the marking scheme can be marked wrong. For π, use the calculator value or 3.142.
- Do not substitute the signs of the charges into Coulomb's law; use magnitudes and find the direction of the force with a positive test charge (like charges repel, unlike attract).

SECTION A TRAPS

Trap: Missing square: using $F \propto \frac{1}{r}$ instead of $\frac{1}{r^2}$. Doubling r gives $\frac{F}{4}$ (not $\frac{F}{2}$); tripling r gives $\frac{F}{9}$.

Trap: Unconverted prefixes: using 3 μC as 3 or 3×10^{-3} C, or 30 cm as 30 m, gives answers wrong by powers of ten; always check the order of magnitude.

Trap: Wrong factor in a combined change: doubling one charge and halving the separation gives 2 × 4 = 8F (not 4F); doubling one charge and tripling r gives $\frac{2F}{9}$.

Trap: Medium: in a medium ε_r divides the force (F decreases); a distractor multiplies by ε_r instead.

TRICKS

Trick: Ratio method: write $\frac{F'}{F} = \frac{Q_1'Q_2'}{Q_1Q_2}\left(\frac{r}{r'}\right)^2$ and substitute the factors directly; no constants are needed.

Trick: Null point between two like charges: it lies nearer the smaller charge, and $\frac{r_1}{r_2} = \sqrt{\frac{Q_1}{Q_2}}$; for +Q and +4Q the ratio of distances is 1 : 2.

13.2 Electric Field

DEFINITIONS & LAWS

- An **electric field** is a region in which an electric force acts on a charged body.
- **Electric field strength** at a point is the force per unit positive charge placed at that point ($E = \frac{F}{q}$); it is a vector quantity whose direction is that of the force on a positive test charge.
- **Electric field line**: a line whose tangent at any point gives the direction of the electric field at that point; the density of the lines is proportional to the field strength.
- Field lines begin on positive charges (or at infinity) and end on negative charges (or at infinity); they never cross, because the field has only one direction at a point.
- The field of a point charge is radial: outward for a positive charge and inward for a negative charge.
- A **uniform field** exists between two oppositely charged parallel plates: the field lines are parallel and equally spaced, and E has the same value at every point (apart from the edges).
- **Superposition**: the resultant field at a point is the vector sum of the fields due to the individual charges.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Electric field strength	$E = \frac{F}{q}$	E = electric field strength (N C^{-1} = V m^{-1}); F = force on the test charge (N); q = test charge (C)
Field of a point charge	$E = \frac{Q}{4\pi\varepsilon_0 r^2} = \frac{kQ}{r^2}$	Point charge, or outside a charged sphere; Q = charge (C); r = distance from the centre (m)
Field between parallel plates	$E = \frac{V}{d}$	V = potential difference (V); d = plate separation (m)
Field and potential gradient	$E = -\frac{dV}{dr}$	$\frac{dV}{dr}$ = potential gradient (V m^{-1}); the field points in the direction of decreasing potential
Force and acceleration of a charge	$F = qE;\ a = \frac{qE}{m}$	F = force (N); a = acceleration (m s^{-2}); q = charge (C); m = mass (kg)
Charge starting from rest	$v^2 = 2as$	Charge starting from rest with uniform acceleration; v = final speed (m s^{-1}); s = distance (m)
Charge entering perpendicular to field	Parabolic path	The path is a parabola, with constant velocity component along the plates
Field near a charged conducting plate	$E = \frac{\sigma}{\varepsilon_0}$	Near a large charged conducting plate; σ = surface charge density (C m^{-2}); ε_0 = permittivity of free space (F m^{-1})

TIPS TO ANSWER

- Calculate the field of each charge separately, give each a direction, then add as vectors; end with the magnitude and direction of the resultant.
- For a charge in a field state that the force is F = qE, which is in the direction of E for a positive charge and opposite to E for a negative charge, then use a = F/m.
- Draw field lines with arrows, with correct start/end on charges, no crossing, and closer spacing where the field is stronger.
- State the direction of E clearly: "away from the positive charge" or "towards the negative charge".
- Electric field strength is a vector: find each field by placing a positive test charge at the point, then add the fields as vectors (draw them). Example: midway between −5 μC and +5 μC, both fields point towards the negative charge, so they add.

SECTION A TRAPS

Trap: Field direction: at the midpoint between +Q and −Q the fields add (same direction, towards −Q) so E = $\frac{2kQ}{r^2}$, not zero; for two equal like charges the fields cancel at the midpoint.

Trap: Inverse proportionality: E = $\frac{kQ}{r^2}$ falls as $\frac{1}{r^2}$; a distractor uses $\frac{1}{r}$. Close to a large plate the field is constant, not dependent on r.

Trap: Electron motion: an electron accelerates opposite to the direction of E; a = $\frac{eE}{m_e}$, using m_e = 9.11×10^{-31} kg and not the mass of a proton.

Trap: Field lines: options claiming that lines may cross, or that the tangent gives the velocity of the charge, are incorrect.

TRICKS

Trick: Null-point test: for unlike charges the field is zero only outside the pair, on the side of the smaller charge; for like charges it is zero between them, nearer the smaller charge.

Trick: A zero potential does not mean zero field, and a zero field does not mean zero potential; keep the two quantities separate.

13.3 Gauss's Law

DEFINITIONS & LAWS

- **Electric flux** through a surface is the product of the electric field strength normal to the surface and the area of the surface, $\Phi = EA\cos\theta$, where θ is the angle between E and the normal to the surface.
- **Gauss's law** states that the total electric flux through any closed surface is equal to the net charge enclosed by the surface divided by ε_0.
- The flux depends only on the net enclosed charge, not on the shape or size of the closed surface, and not on the position of the charge inside the surface.
- A charge outside the closed surface contributes zero net flux, because every field line that enters the surface also leaves it.
- In electrostatic equilibrium the field inside the material of a conductor is zero, so the excess charge lies on the surface of the conductor.
- **Surface charge density** σ is the charge per unit area of a surface, $\sigma = \frac{Q}{A}$.
- Outside a charged sphere the field is the same as if all the charge were concentrated at the centre.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Gauss's law	$\Phi = \oint E \cdot dA = \frac{\Sigma Q}{\varepsilon_0}$	Φ = electric flux (N m^2 C^{-1} = V m); E = electric field strength (N C^{-1}); A = area (m^2); ΣQ = algebraic sum of the enclosed charges (C); ε_0 = permittivity of free space (F m^{-1})
Field of a point charge or sphere	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$	Valid for r ≥ radius R; Q = charge (C); r = distance from the centre (m)
Field of an infinite line charge	$E = \frac{\lambda}{2\pi\varepsilon_0 r}$	λ = charge per unit length (C m^{-1}); r = perpendicular distance from the line (m)
Field of an infinite plane sheet	$E = \frac{\sigma}{2\varepsilon_0}$	σ = charge per unit area (C m^{-2})
Field at a conductor surface	$E = \frac{\sigma}{\varepsilon_0}$	Just outside the surface of a conductor; σ = charge per unit area (C m^{-2})

- $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}.

TIPS TO ANSWER

- State Gauss's law, choose a Gaussian surface with the symmetry of the charge distribution (sphere, coaxial cylinder or pillbox) and justify that E is constant over it and perpendicular to it.
- Write Φ = E × (area of surface) = $\frac{Q_{enclosed}}{\varepsilon_0}$, then solve for E and state its dependence ($\frac{1}{r^2}$, $\frac{1}{r}$ or constant).
- For flux questions add the enclosed charges algebraically with their signs before dividing by ε_0.
- For a conductor state: E = 0 inside, because the enclosed charge for a Gaussian surface in the conductor is zero, so the charge resides on the outer surface.

- To show that Gauss's law is equivalent to Coulomb's law: draw a Gaussian sphere round a point charge Q, apply Gauss's law, use the definition of electric flux ($\Phi = E \times 4\pi r^2$) to get E, then place a charge q and use $E = \frac{F}{q}$ to obtain Coulomb's law.

SECTION A TRAPS

Trap: Surface size: doubling the radius of a sphere around a central charge leaves the flux unchanged; options with $\frac{1}{4}$ or $\frac{1}{2}$ are wrong because they confuse E with Φ.

Trap: Charge outside: a charge not enclosed gives zero net flux however near it is; distractors give $\frac{Q}{\varepsilon_0}$ or $\frac{Q}{2\varepsilon_0}$.

Trap: Sign slip: ΣQ must be algebraic; +3Q, −Q and +2Q give $\frac{4Q}{\varepsilon_0}$, not $\frac{6Q}{\varepsilon_0}$.

Trap: Field forms: the line-charge field is proportional to $\frac{1}{r}$ (not $\frac{1}{r^2}$), and the plane sheet field is independent of r.

TRICKS

Trick: Symmetry fractions: a charge at the centre of a cube gives $\frac{Q}{\varepsilon_0}$ through the whole cube, so $\frac{Q}{6\varepsilon_0}$ through each face; a charge at one corner shared by 8 cubes gives $\frac{Q}{8\varepsilon_0}$ through each cube.

Trick: Total flux through a closed surface never contains r: reject any option such as $\frac{Q}{4\pi\varepsilon_0 r^2}$, which is the field E and not the flux.

13.4 Electric Potential

DEFINITIONS & LAWS

- **Electric potential** at a point is the work done per unit positive charge in bringing a small test charge from infinity to that point ($V = \frac{W}{q}$).
- **Potential difference** between two points is the work done per unit positive charge in moving a test charge from one point to the other.
- **Electric potential energy** of a charge at a point is the work done in bringing the charge from infinity to that point ($U = qV$).

- An **equipotential surface** is a surface on which all points are at the same potential; no work is done in moving a charge along it, and it is perpendicular to the electric field lines.
- The electric field is the negative potential gradient ($E = -\frac{dV}{dr}$): the field points in the direction of decreasing potential, and its magnitude is large where equipotentials are close together.
- Potential is a scalar: the potential due to several charges is the algebraic sum of the individual potentials.
- One **electron-volt** (eV) is the energy gained by an electron when it is accelerated through a potential difference of 1 V; 1 eV = 1.60×10^{-19} J.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Potential of a point charge	$V = \frac{Q}{4\pi\varepsilon_0 r} = \frac{kQ}{r}$	V = electric potential (V); Q = charge (C); r = distance from a point charge, or from the centre of a charged sphere for r ≥ R (m)
Work done by an external force	$W = q(V_B - V_A) = q\Delta V$	Work done by an external force in moving charge q from A to B (J); q = charge (C); V_A, V_B = potentials at A and B (V)
Work done by the field	$W = q(V_A - V_B)$	Work done by the field in moving charge q from A to B (J)
Electric potential energy	$U = qV$	U = potential energy (J); q = charge (C); V = potential (V)
Potential energy of two point charges	$U = \frac{kQ_1Q_2}{r}$	Q_1, Q_2 = point charges (C); r = separation (m); $k = \frac{1}{4\pi\varepsilon_0}$
Field and potential	$E = \frac{V}{d}$; $E = -\frac{dV}{dr}$	$E = \frac{V}{d}$ for a uniform field; $E = -\frac{dV}{dr}$ in general; E = field strength (V m^{-1}); d = separation (m)
Charge accelerated through p.d.	$\frac{1}{2}mv^2 = qV$	Charge accelerated from rest through potential difference V; m = mass (kg); v = speed (m s^{-1}); q = charge (C); V = potential difference (V)
Charged oil drop at rest	$\frac{qV}{d} = mg$	Charged oil drop held at rest between plates; q = charge (C); V = p.d. (V); d = plate separation (m); m = mass (kg); g = acceleration of free fall (m s^{-2})

- 1 eV = 1.60×10^{-19} J.

TIPS TO ANSWER

- State the formula, substitute with signs for each charge ($V = \frac{Q}{4\pi\varepsilon_0 r}$ keeps the sign of Q), add algebraically, and give the answer in V with the sign stated.

- For work done, state who does it: by the field $q(V_A - V_B)$, by an external force $q(V_B - V_A)$; give the sign and unit (J).
- For a drop in balance write $qE = mg$ with $E = \frac{V}{d}$, using the total potential difference between the plates and the separation in metres.
- For equipotentials sketch concentric circles (spheres) round a point charge, with field lines radial and perpendicular to them.
- Potential is a scalar, so substitute the signs of the charges and add algebraically. Example: midway between −5 µC and +5 µC, $V = 0$ although E is not zero.
- Two identical charged spheres touched and then separated share the total charge equally. Spheres joined by a wire exchange charge until their potentials are equal.

SECTION A TRAPS

Trap: Vector addition of potentials: V is a scalar; +4 µC and −4 µC at equal distances give $V = 0$ at the midpoint, though $E \neq 0$ there; a distractor gives $\frac{2kQ}{r}$.

Trap: Sign of work: a positive charge moving from high to low potential has positive work done by the field; the external work is of opposite sign.

Trap: Potential difference between plates: plates at +150 V and −150 V have $\Delta V = 300$ V, not 150 V.

Trap: Direction: potential decreases along the field direction; a distractor says it increases or is undefined away from point charges.

TRICKS

Trick: Zero-V trick: equal and opposite charges at equal distances give $V = 0$; check the sign of each term before adding.

Trick: Distance check: for the centre of a square of side d, each corner is at $\frac{d}{\sqrt{2}}$, so the factor $\sqrt{2}$ appears ($V = 4\sqrt{2}\ \frac{kQ}{d}$ for four equal charges).

Chapter 14

CAPACITORS

14.1 Capacitance

DEFINITIONS & LAWS

- **Capacitance** of a capacitor is the ratio of the magnitude of the charge on either plate to the potential difference between the plates ($C = \frac{Q}{V}$).
- **One farad** is the capacitance of a capacitor that stores a charge of one coulomb when the potential difference across it is one volt ($1\ F = 1\ C\ V^{-1}$).
- The capacitance depends only on the geometry of the capacitor (area, separation, shape) and the dielectric between the plates; it does not depend on the charge stored or the potential difference applied.
- When a capacitor is connected to a battery the potential difference V is fixed; when it is isolated (disconnected) the charge Q remains constant.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Capacitance	$C = \frac{Q}{V}$	C = capacitance (F); Q = charge on one plate (C); V = potential difference across the plates (V)
Capacitance of an isolated sphere	$C = 4\pi\varepsilon_0 R$	Isolated conducting sphere; R = radius (m); ε_0 = permittivity of free space ($F\ m^{-1}$)
Parallel plate capacitor (air)	$C = \frac{\varepsilon_0 A}{d}$	A = area of one plate (m^2); d = plate separation (m)
Parallel plate capacitor (dielectric)	$C = \varepsilon_0 \varepsilon_r \frac{A}{d}$	ε_r = relative permittivity of the dielectric (no unit)
Unit of capacitance	$F = C\ V^{-1} = A^2\ s^4\ kg^{-1}\ m^{-2}$	Typical values are μF (10^{-6} F), nF (10^{-9} F) and pF (10^{-12} F)

- On a graph of Q against V (straight line through the origin) the gradient is C.

TIPS TO ANSWER

- State $C = \frac{Q}{V}$, substitute with units converted (μF, μC, mV) and give the answer in F with the suitable prefix.

- For a change question, state first which quantity is constant: V constant (still connected to battery) or Q constant (isolated), then use $Q = CV$ or $V = \frac{Q}{C}$.
- For a definition write "ratio of charge on one plate to the potential difference across the plates", and do not omit the word ratio.
- For a graph question, label the axes and state that the gradient of the Q–V graph equals the capacitance.
- Define capacitance in words for both marks: the ratio of the charge on either plate to the potential difference between the plates, $C = \frac{Q}{V}$. A formula alone is not enough.

SECTION A TRAPS

Trap: Wrong dependence: options stating that C is proportional to Q or to V; C is constant for a given capacitor.

Trap: Constant quantity: with the battery disconnected, doubling C halves V; with the battery connected, doubling C doubles Q.

Trap: Sphere sharing: two identical spheres sharing charge keep the same capacitance; $C = 4\pi\varepsilon_0 R$ depends on R only.

Trap: Unit slip: a distractor writes the farad with A instead of A^2, or with s^2 instead of s^4; check using $F = C\,V^{-1}$.

TRICKS

Trick: $Q = CV$ is a direct proportion: if V rises by $\frac{20}{12}$, Q rises by the same factor, and C need not be found.

Trick: Inverse relation: for the same charge a larger V means a smaller capacitance, since $C = \frac{Q}{V}$.

14.2 Parallel Plate Capacitors

DEFINITIONS & LAWS

- For a parallel plate capacitor the capacitance is directly proportional to the area of the plates and inversely proportional to their separation.
- The electric field between the plates is uniform, $E = \frac{V}{d} = \frac{\sigma}{\varepsilon_0}$, directed from the positive plate to the negative plate; the edge effects are neglected when d is much smaller than the plate dimensions.

- If the battery stays connected V is constant, so Q = CV changes when d or A changes; if the capacitor is isolated Q is constant.
- A slab of dielectric that partly fills the gap in layers (parallel to the plates) forms capacitors in series; a slab covering part of the area forms capacitors in parallel.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Parallel plate capacitance (air)	$C = \frac{\varepsilon_0 A}{d}$	C = capacitance (F); A = area of one plate (m^2); d = plate separation (m); $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}
Capacitance with dielectric	$C = \varepsilon_0\varepsilon_r \frac{A}{d}$	Dielectric completely filling the gap; ε_r = relative permittivity (no unit)
Field between the plates	$E = \frac{V}{d} = \frac{Q}{\varepsilon_0 A}$	E = electric field strength (V m^{-1}); V = potential difference (V); Q = charge (C)
Layers in series	$\frac{1}{C} = \frac{d_1}{\varepsilon_0 A} + \frac{d_2}{\varepsilon_0\varepsilon_r A}$	d_1, d_2 = layer thicknesses (m); ε_r = relative permittivity of the second layer
Stored energy	$U = \frac{1}{2}CV^2$	U = energy stored (J); C = capacitance (F); V = potential difference (V)

TIPS TO ANSWER

- Write $C = \frac{\varepsilon_0 A}{d}$, convert cm^2 to m^2 ($\times 10^{-4}$) and mm to m ($\times 10^{-3}$) before substituting, then answer in F or pF with 3 s.f.
- For change questions use $C \propto \frac{A}{d}$, state what is constant (V or Q) and then use $Q = CV$ or $V = \frac{Q}{C}$.
- For a layered capacitor find each layer's capacitance ($C = \varepsilon_0\varepsilon_r \frac{A}{d_{layer}}$) and combine in series.
- For field strength use $E = \frac{\sigma}{\varepsilon_0} = \frac{Q}{\varepsilon_0 A}$, or $E = \frac{V}{d}$, and state the units V m^{-1}.

SECTION A TRAPS

Trap: Inverted dependence: C increases when d decreases or A increases; options with C increased by increasing d or removing the dielectric are wrong.

Trap: Battery connected: when d increases, C decreases and Q = CV decreases; when d doubles Q is halved, not doubled.

Trap: Area unit: 6.0 cm^2 = 6.0×10^{-4} m^2 (not 6.0×10^{-2} m^2); a missing conversion gives an answer wrong by a factor of 10^2 or 10^3.

Trap: Combined change: halving d and doubling A multiplies C by 4, not 2.

TRICKS

Trick: Ratio method: $\frac{C'}{C} = \frac{A'}{A}\frac{d}{d'}$; the constants ε_0 cancel.

Trick: Layered gap: each layer of thickness $\frac{d}{2}$ has twice the air capacitance, so in series the value is quickly found as $\frac{2C_0 \times 6C_0}{2C_0 + 6C_0} = 1.5C_0$ for $\varepsilon_r = 3$.

14.3 Dielectrics

DEFINITIONS & LAWS

- A **dielectric** is an insulating material which becomes polarised in an electric field.
- **Relative permittivity (dielectric constant)** ε_r is the ratio of the capacitance of a capacitor with the dielectric between the plates to its capacitance when there is a vacuum between them ($\varepsilon_r = \frac{C}{C_0} = \frac{\varepsilon}{\varepsilon_0}$).
- **Polarisation:** in the external field the molecules of the dielectric become aligned (polar molecules) or stretched (non-polar molecules), so that opposite induced charges appear on the faces of the dielectric.
- The induced charges produce an electric field opposite in direction to the applied field, so the net field in the dielectric is reduced ($E = \frac{E_0}{\varepsilon_r}$), the potential difference between the plates falls for the same charge and so the capacitance increases.
- **Dielectric breakdown** occurs when the field exceeds the dielectric strength of the material and the insulator begins to conduct.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Capacitance with dielectric	$C = \varepsilon_r C_0 = \varepsilon_0 \varepsilon_r \frac{A}{d}$	C = capacitance (F); C_0 = capacitance without dielectric (F); ε_r = relative permittivity (no unit); A = plate area (m^2); d = plate separation (m)
Isolated capacitor (Q constant): field and p.d.	$E = \frac{E_0}{\varepsilon_r}$; $V = \frac{V_0}{\varepsilon_r}$	Capacitor isolated at constant Q; E_0, V_0 = field strength ($V\ m^{-1}$) and p.d. (V) without dielectric

Quantity / law	Formula	Symbols, units & conditions
Connected to battery (V constant)	$Q = \varepsilon_r Q_0$; $U = \frac{1}{2}\varepsilon_r C_0 V^2 = \varepsilon_r U_0$; $E = \frac{V}{d}$ unchanged	Battery connected, V constant; Q_0 = charge without dielectric (C); U_0 = energy without dielectric (J); U = energy (J)
Isolated capacitor: stored energy	$U = \frac{Q^2}{2C} = \frac{U_0}{\varepsilon_r}$	Isolated, Q constant; U = energy (J); Q = charge (C); C = capacitance (F)
Maximum voltage	$V_{max} = E_{max}\, d$	V_{max} = maximum safe voltage (V); E_{max} = breakdown field strength (V m^{-1}); d = plate separation (m)

TIPS TO ANSWER

- Explain a capacitance increase in three points: the dielectric is polarised, the induced charges produce a field opposing the original field, the net field and the potential difference fall for the same charge, so $C = \frac{Q}{V}$ increases.
- State at the start whether the capacitor is connected to the battery (V constant) or isolated (Q constant); the answer depends on it.
- For numerical work use $C' = \varepsilon_r C$, then $Q = C'V$ (connected) or $V' = \frac{Q}{C'}$ (isolated) and give the answer with unit.
- Use the word "partially" for the cancelling effect: the net field is reduced, not zero.
- Explain why a dielectric increases the capacitance in sequence: (1) positive nuclei are attracted to the negative plate and electrons to the positive plate; (2) dipoles form and an opposing field is set up; (3) the p.d. between the plates decreases while Q stays constant; (4) from $C = \frac{Q}{V}$, C increases. Do not jump to the last point.
- Isolated capacitor (Q constant): inserting a dielectric decreases V. Capacitor kept connected to a battery (V constant): inserting a dielectric increases Q.

SECTION A TRAPS

Trap: Isolated vs connected: inserting a dielectric into an isolated capacitor reduces V, E and U by ε_r; with the battery connected Q and U increase by ε_r and V and E stay the same.

Trap: Factor error: energy changes by ε_r, not ε_r^2; a ratio of 1 : 25 or 25 : 1 for $\varepsilon_r = 5.0$ is wrong.

Trap: Complete cancellation: options saying the net field in the dielectric is zero, or that the polarisation field reinforces the external field, are wrong.

Trap: Breakdown: V_{max} depends on E_{max} and d, not on ε_r; doubling E_{max} doubles V_{max}.

TRICKS

Trick: Quick table: connected: V same, E same, $Q \times \varepsilon_r$, $U \times \varepsilon_r$. Isolated: Q same, $V \div \varepsilon_r$, $E \div \varepsilon_r$, $U \div \varepsilon_r$.

Trick: Use $U = \frac{1}{2}QV$: if Q is constant, U follows V; if V is constant, U follows Q.

14.4 Capacitors in Series and in Parallel

DEFINITIONS & LAWS

- In a **parallel** combination all capacitors have the same potential difference across them, and the total charge is the sum of the charges on the capacitors.
- In a **series** combination every capacitor carries the same charge, and the total potential difference is the sum of the potential differences across the capacitors.
- The equivalent capacitance in parallel is the sum of the capacitances; in series the reciprocal of the equivalent capacitance is the sum of the reciprocals.
- In series the potential difference across a capacitor is inversely proportional to its capacitance, so the smaller capacitor has the larger potential difference.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Capacitors in parallel	$C = C_1 + C_2 + C_3 + \ldots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
Capacitors in series	$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \ldots$	C = total capacitance (F); C_1, C_2, C_3 = individual capacitances (F)
Two capacitors in series	$C = \frac{C_1 C_2}{C_1 + C_2}$	C_1, C_2 = capacitances (F)
Series charge and voltage sharing	$Q = CV$; $V_1 = \frac{VC_2}{C_1 + C_2}$; $V_2 = \frac{VC_1}{C_1 + C_2}$	Q = charge on each capacitor (C); V = total p.d. (V); V_1, V_2 = p.d. across C_1 and C_2 (V)
n identical capacitors	Parallel: nC; series: $\frac{C}{n}$	n = number of identical capacitors, each of capacitance C (F)
Total stored energy	$U_{total} = \frac{1}{2}C_{total}V^2$	U_{total} = energy of the whole arrangement (J); C_{total} = total capacitance (F); V = p.d. across the arrangement (V)

TIPS TO ANSWER

- Reduce the network step by step: combine each series or parallel group, redraw, and finish with a single equivalent capacitance.
- For voltage division use $Q = C_{total} \times V$ (same for each series capacitor) and then $V_1 = \frac{Q}{C_1}$.
- Write each stage with units (μF) and give the final value to 3 s.f.
- State the rule used at each step: "same charge in series" or "same potential difference in parallel".
- Decide series or parallel from how the plates are joined, not from how the diagram looks. A charged capacitor connected to an uncharged one is in parallel with it: total charge is conserved, so $V = \frac{C_1 V_0}{C_1 + C_2}$. Example: C_1 charged to 100 V, final p.d. 60 V: $100C_1 = 60(C_1 + C_2)$, so $\frac{C_1}{C_2} = 1.5$.

SECTION A TRAPS

Trap: Inverted rules: the formulae for capacitors are opposite to those for resistors; a distractor adds the capacitances in series.

Trap: Series bound: the series equivalent must be smaller than the smallest capacitor; 4 μF and 6 μF in series give 2.4 μF, whereas 10 μF (the sum) or 24 μF (the product) is wrong.

Trap: Voltage division: in series the smaller capacitor has the larger share: 3 μF and 6 μF across 12 V give 8 V and 4 V, not 4 V and 8 V.

Trap: Charge in a mixed network: the charge on the single series capacitor equals the sum of the charges on the two parallel capacitors, not equal to each one.

TRICKS

Trick: Bounds: parallel gives more than the largest value; series gives less than the smallest; use this to eliminate options.

Trick: Identical capacitors C: the possible equivalents are 3C, $\frac{C}{3}$, $\frac{3C}{2}$ and $\frac{2C}{3}$ only.

14.5 Energy Stored in a Charged Capacitor

DEFINITIONS & LAWS

- The **energy stored in a capacitor** is the work done in charging it, equal to the area under the graph of potential difference against charge.
- **Energy density** of the electric field is the energy stored per unit volume of the field between the plates.

- When a capacitor is charged from a battery through a resistor, the battery supplies energy QV, of which ½QV is stored in the capacitor and ½QV is dissipated as heat in the resistor, independent of R.
- When a charged capacitor is connected to an uncharged capacitor the charge is conserved but the total energy decreases; the energy lost is dissipated as heat (and radiation).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Energy stored in a capacitor	$U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{Q^2}{2C}$	U = energy (J); Q = charge (C); V = potential difference (V); C = capacitance (F)
Energy density	$u = U/(Ad) = \frac{1}{2}\varepsilon_0 E^2$	u = energy density (J m^{-3}); A = plate area (m^2); d = plate separation (m); E = field strength (V m^{-1}); ε_0 = permittivity of free space (F m^{-1})
Energy density in a dielectric	$u = \frac{1}{2}\varepsilon_0\varepsilon_r E^2$	ε_r = relative permittivity (no unit)
Work done by the battery	$W = QV = CV^2$	W = work done by the battery (J); Q = charge (C); V = p.d. (V); C = capacitance (F)
Parallel sharing	$V = \frac{Q_{total}}{C_1 + C_2}$	V = common p.d. after sharing (V); Q_{total} = total charge (C); C_1, C_2 = capacitances (F)

- $\varepsilon_0 = 8.85\times10^{-12}$ F m^{-1}.

TIPS TO ANSWER

- For the energy stored write $U = \frac{1}{2}CV^2$ (choose the form matching the data), substitute with SI units, and give the answer in J with 3 s.f.
- For the V–Q graph, sketch a straight line through the origin and shade the triangle, showing that the area ½QV equals the energy.
- For energy lost on connecting capacitors find the charge conserved, the final V, the final energy and subtract it from the initial energy.
- Explain a loss as "energy is dissipated as heat in the connecting wires/resistor".
- Choose the energy formula from what stays constant: battery connected (V constant) → $U = \frac{1}{2}CV^2$; battery removed (Q constant) → $U = \frac{Q^2}{2C}$.
- Example: after discharging through R for time t, $Q = 0.37Q_0$, so $\frac{U'}{U} = (0.37)^2 = 0.137$: 13.7% of the energy remains.

SECTION A TRAPS

Trap: Missing ½: U = QV is the energy supplied by the battery, not the energy stored.

Trap: Wrong power: $U \propto V^2$ (so $3V_0$ gives $9E_0$, not $3E_0$) and $U \propto Q^2$ at fixed C.

Trap: Energy conservation: two equal capacitors, one charged, then connected: the final energy is half the initial, not equal to it.

Trap: Sum of energies: the total energy of two separate capacitors is the sum of each $\frac{1}{2}CV^2$, not $\frac{1}{2}(C_1 + C_2)(V_1 + V_2)^2$.

TRICKS

Trick: Choose the form of U: $\frac{1}{2}CV^2$ at constant V, $\frac{Q^2}{2C}$ at constant Q, and $\frac{1}{2}QV$ if both are known.

Trick: Fixed capacitor: ratio method, $\frac{U'}{U} = (\frac{V'}{V})^2$ or $(\frac{Q'}{Q})^2$.

14.6 Charging and Discharging of a Capacitor

DEFINITIONS & LAWS

- The **time constant** $\tau = RC$ is the time taken for the charge on a discharging capacitor to fall to $\frac{1}{e}$ (about 37%) of its initial value; for charging it is the time for the charge to reach 63% $(1 - \frac{1}{e})$ of its final value.
- During discharge the charge, potential difference and current decay exponentially with time; during charging the current decays exponentially while the charge and potential difference rise exponentially towards their final values.
- At the start of charging the capacitor has no potential difference, so the initial current is $I_0 = \frac{V_0}{R}$; when the capacitor is fully charged the current is zero and the potential difference across it equals the EMF.
- The time constant is increased by increasing R or C; the final charge depends on C and V_0 only.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Discharging a capacitor	$Q = Q_0e^{-t/RC}$; $V = V_0e^{-t/RC}$; $I = I_0e^{-t/RC}$	Q, V, I = charge (C), p.d. (V), current (A) at time t (s); Q_0, V_0, I_0 = initial values; $I_0 = \frac{V_0}{R}$; R = resistance (Ω); C = capacitance (F)
Charging a capacitor	$Q = Q_0(1 - e^{-t/RC})$; $V = V_0(1 - e^{-t/RC})$; $I = I_0e^{-t/RC}$	$Q_0 = CV_0$ = final charge (C); V_0 = final p.d. (V); I_0 = initial current (A); R = resistance (Ω); C = capacitance (F); t = time (s)

Quantity / law	Formula	Symbols, units & conditions
Time constant	$\tau = RC$	τ = time constant (s); R = resistance (Ω); C = capacitance (F); $e^{-1} = 0.368$, $e^{-2} = 0.135$
Time for charge to halve	$t_{½} = RC \ln 2 = 0.693RC$	$t_{½}$ = time for the charge to halve (s); R = resistance (Ω); C = capacitance (F)
Logarithmic graph (discharge)	Gradient of ln Q against t = $-\frac{1}{RC}$	ln Q against t is a straight line of gradient $-\frac{1}{RC}$ (discharge)

TIPS TO ANSWER

- Sketch the graph with labelled axes: Q (or V, I) against t, exponential decay from the initial value that tends to zero, or growth from zero to the final value $Q_0 = CV_0$; mark τ.
- Write the equation, substitute $\frac{t}{RC}$ with RC in seconds (kΩ × μF = ms), evaluate the exponential and give the answer with 3 s.f.
- For a time, rearrange with natural logarithms, e.g. $t = -RC \ln \frac{V}{V_0}$ for discharge or $t = -RC \ln(1 - \frac{V}{V_0})$ for charging.
- State that the exponential never reaches zero in a finite time, and that about 5τ is taken as practically complete.

SECTION A TRAPS

Trap: Meaning of τ: after $t = \tau$ the charge in discharge is $\frac{1}{e}$ (37%) of the initial charge, not zero or half; in charging the capacitor reaches 63%, not 37%.

Trap: Dependence on R: doubling R doubles the time; the initial current $\frac{V_0}{R}$ decreases but the final charge CV_0 is unchanged.

Trap: Two time constants: $\frac{Q}{Q_0} = e^{-2} = 0.135$ (about 14%); a distractor gives 0.37 or 0.74.

Trap: Half-life of the current: I falls to $\frac{I_0}{2}$ at $t = RC \ln 2$ (0.693RC), not $\frac{RC}{2}$ or RC.

TRICKS

Trick: Percentages to recall: charging at τ = 63%, at 2τ = 86%; discharging at τ = 37%, at 2τ = 14%.

> **Trick:** Unit check: kΩ × μF gives ms, kΩ × mF gives s, MΩ × μF gives s.

Chapter 15

ELECTRIC CURRENT

15.1 Conduction of Electricity

DEFINITIONS & LAWS

- **Electric current** is the rate of flow of charge through a cross-section of a conductor.
- **Conduction in a metal** is due to the free (delocalised) electrons, which have large random thermal velocities in all directions; an applied p.d. sets up an electric field that superimposes a small net drift velocity opposite to the field.
- **Conventional current** is taken in the direction of flow of positive charge, which is opposite to the direction of drift of the electrons.
- **Explain why the resistance of a metal increases with temperature:** the lattice ions vibrate with greater amplitude, so the free electrons collide with them more frequently; the mean free time between collisions decreases while the free electron density is almost constant.
- **Explain why the conductivity of a semiconductor increases with temperature:** more electrons are released from the covalent bonds, producing more electron-hole pairs; the increase in the number of charge carriers outweighs the increased lattice scattering.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Electric current	$I = \frac{Q}{t}$	I = current (A, where A = C s^{-1}); Q = charge passing a point (C); t = time (s)
Charge and number of electrons	$Q = Ne$	N = number of electrons; e = 1.60×10^{-19} C
Current and drift velocity	$I = nAev_d$	n = free electron density (m^{-3}); A = cross-sectional area (m^2); v_d = drift velocity (m s^{-1})

- Typical values: drift velocity ~10^{-4} m s^{-1}; random thermal speed ~10^5 to 10^6 m s^{-1}.

TIPS TO ANSWER

- For 'explain' questions, state the three linked points: free electrons, collisions with lattice ions, and the effect on the mean free time / drift velocity.
- State that the electrons drift in the direction opposite to the electric field and to the conventional current.

- For charge-counting problems: $Q = It$, then $N = \frac{Q}{e}$, with time converted to seconds and the answer in standard form.
- When contrasting metals and semiconductors, compare the change in carrier density with the change in scattering.

SECTION A TRAPS

Trap: The resistance of a metal increases because the number of free electrons falls with temperature. The free electron density is nearly constant; it is the collision rate that increases.

Trap: Forgetting to convert minutes to seconds in $Q = It$, giving an answer 60 times too small.

Trap: Stating that the free electrons move in the direction of the conventional current; they drift in the opposite direction.

TRICKS

Trick: Metal: n constant, collisions increase, so R increases. Semiconductor: n increases faster than the scattering, so R decreases. Eliminate any option that changes the charge e of the carrier.

15.2 Drift Velocity

DEFINITIONS & LAWS

- **Drift velocity** is the **mean** velocity of the free electrons along the conductor due to the applied electric field, superimposed on their random thermal motion.
- **Why a lamp lights almost instantaneously:** the electric field is established along the whole circuit at a speed close to that of light, so the free electrons throughout the wire, including those in the filament, begin to drift at once.
- **In a steady current** the drift velocity is the same at every point of a uniform wire; in a wire of varying cross-section it is greater where the area is smaller.
- **In a semiconductor** both electrons and holes contribute to the current; they drift in opposite directions but their contributions to the conventional current add.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Current and drift velocity	$I = nAev_d$	n = number of free charge carriers per unit volume (m^{-3}); A = cross-sectional area (m^2); e = 1.60×10^{-19} C; v_d = drift velocity (m s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Drift velocity	$v_d = \frac{I}{nAe}$	I = current (A)
Proportionality of drift velocity	$v_d \propto I$; $v_d \propto \frac{1}{nA}$	Each holds with the other quantities constant
Circular cross-sectional area	$A = \pi r^2 = \frac{\pi d^2}{4}$	r = radius (m); d = diameter (m); for a circular cross-section
Current in a semiconductor	$I = neA(v_e + v_h)$	v_e = electron drift velocity (m s^{-1}); v_h = hole drift velocity (m s^{-1})

TIPS TO ANSWER

- State the formula I = nAevd, then rearrange for the unknown before substituting, with every quantity in SI units.
- Find A from the diameter first: $A = \frac{\pi d^2}{4}$, with d converted to metres (1 mm = 10^{-3} m).
- For ratio questions write $v_d \propto \frac{1}{A}$ (same material and current) and substitute the area ratio.
- Give the answer in m s^{-1} in standard form to 3 s.f.; the typical value is of order 10^{-4} m s^{-1}, which serves as a check.
- Define drift velocity as the **mean** velocity of the free electrons; leaving out "mean" loses the mark.
- In $\sigma = \frac{ne^2t}{m}$, t is the mean free time (mean time between collisions), as in A-level reference books, not the time for an electron to travel from one end of the wire to the other.

SECTION A TRAPS

Trap: Using the radius as the diameter, or the reverse, in $A = \frac{\pi d^2}{4}$; this gives an error of a factor of 4.

Trap: Halving the diameter gives double the drift velocity. The area falls to one quarter, so vd becomes 4 times larger.

Trap: Believing that the electrons themselves travel at nearly the speed of light; only the electric field is established at that speed.

Trap: Writing $v_d \propto n$; for fixed I and A, vd is inversely proportional to n.

TRICKS

Trick: Same I, same material: vd $\propto \frac{1}{A} \propto \frac{1}{d^2}$. Halve d, vd × 4; double d, vd ÷ 4.

Trick: A numerical check: copper wires carrying a few amperes give vd between 10^{-5} and 10^{-3} m s^{-1}; an answer far from this range signals a wrong power of ten.

15.3 Current Density

DEFINITIONS & LAWS

- **Current density** is the current per unit cross-sectional area perpendicular to the direction of the current at that point, $J = \frac{I}{A}$.
- **Current density** is a vector quantity with the direction of the conventional current.
- **In a steady current** the current I is the same at every cross-section of a conductor, but the current density is greater where the cross-sectional area is smaller.
- **Ohm's law (microscopic form):** the current density in a conductor is directly proportional to the electric field strength, $J = \sigma E$, at constant temperature.
- **Analogy:** electric current corresponds to the rate of heat flow, and potential difference to temperature difference; conductivity corresponds to thermal conductivity.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Current density	$J = \frac{I}{A}$	J = current density (A m^{-2}); I = current (A); A = cross-sectional area (m^2)
Current density and drift velocity	$J = nev_d$	n = free electron density (m^{-3}); e = 1.60×10^{-19} C; v_d = drift velocity (m s^{-1})
Current density and field	$J = \sigma E$	σ = electrical conductivity (Ω^{-1} m^{-1} or S m^{-1}); E = electric field strength (V m^{-1})
Field in a uniform wire	$E = \frac{V}{L}$	V = p.d. across the wire (V); L = length of wire (m); for a uniform wire

TIPS TO ANSWER

- Convert the area to m^2 before substituting: 1 mm^2 = 10^{-6} m^2, 1 cm^2 = 10^{-4} m^2.
- Write $J = \frac{I}{A}$, substitute with units, then state the answer in A m^{-2} in standard form.
- For a circular wire, calculate $A = \pi r^2$ or $\frac{\pi d^2}{4}$ first and show this step.

- For tapering conductors state that I is constant while $J \propto \frac{1}{A}$, so J is largest at the narrowest section.

SECTION A TRAPS

Trap: Current density is the same at all points because the current is the same; the area differs, so $J = \frac{I}{A}$ differs.

Trap: Mixing mm^2 and m^2, for example 10^{-3} instead of 10^{-6} for mm^2 to m^2; the answer is then wrong by a factor of 1000.

Trap: Area reduced to one third gives J three times larger, not nine times; J is inversely proportional to A, not to A^2.

TRICKS

Trick: $J \propto \frac{1}{A}$ at constant current: area ÷ 3 gives J × 3.

Trick: Use units to check: A m^{-2} requires the area in m^2; for a 1 mm^2 wire carrying 1 A, J = 10^6 A m^{-2}, a useful benchmark.

15.4 Electric Conductivity and Resistivity

DEFINITIONS & LAWS

- **Resistivity** of a material is the resistance of a wire of that material of unit length and unit cross-sectional area, $\rho = \frac{RA}{L}$ (unit: Ω m).
- **Conductivity** is the reciprocal of resistivity, $\sigma = \frac{1}{\rho}$ (unit: Ω^{-1} m^{-1} or S m^{-1}); both are properties of the material, not of its size or shape.
- **Conductance** is the reciprocal of resistance, $G = \frac{1}{R}$ (unit: siemens, S = Ω^{-1}).
- **Ohm's law**: the steady current in a metallic conductor is directly proportional to the potential difference across it, provided the physical conditions (such as temperature) remain constant.
- A **superconductor** is a material whose resistivity becomes zero at and below its critical temperature T_c; **superconductivity** is this phenomenon of zero resistivity below the critical temperature. Uses: strong electromagnets (MRI, particle accelerators), maglev trains and lossless power transmission.
- **Resistance** is the ratio of the potential difference across a conductor to the current through it, $R = \frac{V}{I}$.

- **Variation with temperature:** for a metal the resistivity increases with temperature because lattice vibrations increase the collision rate; for a semiconductor the resistivity decreases with temperature because the number of charge carriers increases rapidly.
- **Stretching a wire:** at constant volume, if the length is multiplied by k, the area is divided by k, so the resistance is multiplied by k^2.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Resistance of a conductor	$R = \frac{\rho L}{A}$	R = resistance (Ω); ρ = resistivity (Ω m); L = length (m); A = cross-sectional area (m^2)
Conductivity	$\sigma = \frac{1}{\rho} = \frac{L}{RA}$	σ = conductivity ($\Omega^{-1}\ m^{-1}$ or $S\ m^{-1}$)
Resistance at constant volume	$V = AL;\ R \propto L^2$ (or $R \propto \frac{1}{A^2}$)	V = volume (m^3), constant; L = length (m); A = cross-sectional area (m^2)
Relaxation time	$\tau = \frac{m}{ne^2\rho}$	τ = relaxation time (s); m = electron mass (9.11×10^{-31} kg); n = free electron density (m^{-3}); e = 1.60×10^{-19} C; ρ = resistivity (Ω m)
Power dissipated in a resistor	$P = I^2R = \frac{V^2}{R}$	P = power (W); I = current (A); V = p.d. (V); R = resistance (Ω); combine with $R = \frac{\rho L}{A}$ in heater-coil problems

TIPS TO ANSWER

- State $R = \frac{\rho L}{A}$, convert the diameter to metres and calculate $A = \frac{\pi d^2}{4}$ before substituting.
- For ratio problems write each resistance as $\frac{\rho L}{A}$, cancel the common quantities and substitute the length and area ratios.
- For a reshaped wire state explicitly that volume is constant, so A changes when L changes.
- State the final answer in Ω (or $\Omega^{-1}\ m^{-1}$ for σ) to 3 s.f. with the unit.
- Use $\sigma = \frac{ne^2t}{m}$ to explain: conductor vs semiconductor (much larger n in a conductor); conductor heated: atoms vibrate with larger amplitude, the mean time between collisions t decreases, σ decreases and ρ increases; semiconductor heated: n increases, σ increases and ρ decreases.
- Resistivity depends only on the material (and temperature): cutting a wire to half its length halves R, but ρ is unchanged.

SECTION A TRAPS

Trap: Stretching to twice the length gives 2R. The area also halves at constant volume, so the resistance becomes 4R.

Trap: Using cm^2 as m^2: 0.040 cm^2 is 4.0×10^{-6} m^2, not 4.0×10^{-4} m^2.

Trap: Using the diameter as the radius in $A = \pi r^2$, which gives A four times too large and R four times too small.

Trap: Treating conductivity as dependent on the length and area of the sample; it is a property of the material only.

TRICKS

Trick: Two materials with identical dimensions: $\sigma \propto \frac{1}{\rho}$, so doubling ρ halves σ.

Trick: Same material: $R \propto \frac{L}{A} \propto \frac{L}{d^2}$. Length × 2 with area ÷ 2 gives R × 4; length × 2 with diameter ÷ 2 gives R × 2 × 4 = 8 times. Check whether the question gives the area or the diameter.

Chapter 16

DIRECT CURRENT CIRCUITS

16.1 Internal Resistance

DEFINITIONS & LAWS

- **Electromotive force (e.m.f.)** of a source is the energy converted from other forms into electrical energy per unit charge passing through the source; it equals the terminal p.d. when no current flows.
- **Internal resistance** of a source is the resistance to the flow of current within the source itself, arising from the materials of the cell.
- **Potential difference** between two points in a circuit is the electrical energy converted to other forms of energy when a unit charge passes from one point to the other.
- A **shunt** is a resistor of very low resistance connected in parallel with a galvanometer to divert most of the current, converting it into an ammeter.
- A **multiplier** is a resistor of very high resistance connected in series with a galvanometer, converting it into a voltmeter.
- **Terminal potential difference** is the p.d. across the terminals of a source when it delivers current; it is less than the e.m.f. by the 'lost volts' Ir.
- **Maximum power theorem:** the power delivered to the external resistor is maximum when the external resistance equals the internal resistance of the source, R = r.
- **Cells in series** have a total e.m.f. and total internal resistance equal to the sums of the individual values (for cells in the same sense).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
E.m.f. equation	$E = V + Ir = I(R + r)$	E = e.m.f. (V); V = terminal p.d. (V); I = current (A); r = internal resistance (Ω); R = external resistance (Ω)
Terminal p.d.	$V = E - Ir$	A graph of V against I is a straight line with intercept E and gradient −r
Internal resistance	$r = \frac{\Delta V}{\Delta I}$; $r = \frac{E - V}{I}$	$\frac{\Delta V}{\Delta I}$ from two readings of the terminal p.d. (magnitude of the gradient of the V–I graph)
Power in external resistor	$P = I^2R = \frac{E^2R}{(R + r)^2}$	P = power in the external resistor (W)
Maximum power transfer	$R = r$; $P_{max} = \frac{E^2}{4r}$	Power in the external resistor is a maximum when R = r

Quantity / law	Formula	Symbols, units & conditions
Power dissipated in the source	$P_{int} = I^2r$	P_{int} = power dissipated in the internal resistance (W)
Total power supplied	$P = EI$	P = total power supplied by the source (W)

TIPS TO ANSWER

- State $E = V + Ir$ and substitute; find I first from $I = \frac{E}{R + r}$ when the circuit values are given.
- For two readings, write two simultaneous equations $E = V_1 + I_1r$ and $E = V_2 + I_2r$, subtract to find r, then find E.
- For a V-I graph state: intercept on the V-axis = E, gradient = −r.
- Describe terminal p.d. as less than e.m.f. because of the p.d. across the internal resistance when a current flows.
- Simple d.c. circuits can be simplified: read the words given, redraw the circuit by following the current from the positive terminal, then combine series and parallel parts. Use Kirchhoff's laws only for complex circuits that cannot be simplified.
- Maximum power is delivered to the external resistor when R = r. In an MCQ, check first for a short circuit before calculating every option.
- An ammeter has a very low resistance (shunt in parallel); a voltmeter has a very high resistance (multiplier in series). Shunt: $V_{shunt} = V_{meter}$; multiplier: same current, $V = V_{meter} + V_{multiplier}$.

SECTION A TRAPS

Trap: Taking the terminal p.d. as equal to the e.m.f. when the cell is delivering current; V = E only on open circuit.

Trap: Defining internal resistance as $\frac{E}{I}$ or $\frac{V}{I}$; these are the total and external resistances, respectively.

Trap: Adding only the external resistance when finding I, ignoring r; the correct total is R + r.

Trap: Forgetting that a series battery of N cells also has N times the internal resistance.

TRICKS

Trick: A voltmeter reading before the switch is closed gives E; the reading afterwards gives V. Then $r = \frac{E - V}{I}$.

> **Trick:** For maximum power, the external and internal resistances are equal, so half of the total power is dissipated in the source (efficiency 50%).

16.2 Kirchhoff's Laws

DEFINITIONS & LAWS

- **Kirchhoff's first law:** the sum of the currents entering a junction equals the sum of the currents leaving the junction. It is based on the conservation of charge.
- **Kirchhoff's second law:** in any closed loop the algebraic sum of the e.m.f.s equals the algebraic sum of the products IR (potential drops) in the loop. It is based on the conservation of energy.
- **Sign convention:** choose a direction for each current and for the loop; an e.m.f. is positive if the loop direction passes through the cell from the negative to the positive terminal, and IR is positive if the loop direction is the same as the current direction.
- **Opposing cells** in a loop give a net e.m.f. equal to the difference of the e.m.f.s.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Kirchhoff's first law (junction rule)	$\Sigma I_{in} = \Sigma I_{out}$	I = current (A); applies at a junction
Kirchhoff's second law (loop rule)	$\Sigma E = \Sigma IR$	E = e.m.f. (V); I = current (A); R = resistance (Ω); applies around a closed loop
Resistors in series	$R = R_1 + R_2 + \ldots$	R = total resistance (Ω)
Resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$	R = total resistance (Ω)
Resistor with a voltmeter across it	$\frac{RR_V}{R + R_V}$	R_V = finite resistance of the voltmeter (Ω); R = resistor (Ω); combined resistance when the voltmeter is connected across R

TIPS TO ANSWER

- Draw the circuit with labelled currents and a chosen loop direction before writing the equations.
- Write the junction equation and one loop equation per independent loop, then solve the simultaneous equations.
- State the direction of each current found; a negative answer means the current flows opposite to the direction assumed.
- For a real voltmeter, replace the resistor and voltmeter by their parallel equivalent before applying the potential divider rule.

- You may choose any direction for each current and any loops; a correct method gives the same final answer. Write the first law (junction) and second law (loop) equations clearly, as each earns marks.
- To find the potential at a point, start from a point of known potential (e.g. earthed = 0 V) and add or subtract the e.m.f.s and IR drops along the path.

SECTION A TRAPS

Trap: Adding the e.m.f.s of opposing cells; the net e.m.f. is the difference, for example 12 V − 4.0 V = 8.0 V.

Trap: Ignoring the voltmeter resistance in a high-resistance circuit; a voltmeter of finite resistance reduces the p.d. it measures.

Trap: Writing the junction law with all currents on the same side but with mixed signs; check that the total entering equals the total leaving.

Trap: Quoting conservation of charge for the loop law; the loop law comes from conservation of energy.

TRICKS

Trick: At a junction, total in minus known out gives the unknown current, with direction determined by the sign of the result.

Trick: A loop with only two cells and resistors in series: $I = \frac{\text{net e.m.f.}}{\text{total resistance}}$; first decide which cells oppose.

16.3 Potential Divider

DEFINITIONS & LAWS

- **A potential divider** is a circuit in which two or more resistors in series are connected across a supply so that the supply voltage is shared in proportion to the resistances; the output is taken across one of them.
- **Output voltage** is directly proportional to the resistance across which it is taken, so the larger resistor receives the larger share.
- **Loading effect:** a load connected in parallel with the output resistor reduces the effective resistance of that arm, so the output p.d. decreases below the no-load value.
- **Sensors:** the resistance of a thermistor (NTC) decreases as temperature increases, and the resistance of an LDR decreases as light intensity increases; the p.d. across the sensor in a divider therefore decreases.

- **A rheostat as a potential divider** has its whole track across the supply, which allows the output to vary continuously from zero to the supply p.d.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Potential divider output	$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$	V_{out} = output p.d. (V); V_{in} = input p.d. (V); R_2 = resistor across which the output is taken (Ω); R_1 = other resistor (Ω)
Ratio of output to input	$\frac{V_{out}}{V_{in}} = \frac{R_2}{R_1 + R_2}$	Ratio is dimensionless
Voltage ratio in series	$V_1 : V_2 = R_1 : R_2$	For resistors in series
Divider with a load	$R_2 \rightarrow \frac{R_2 R_L}{R_2 + R_L}$	R_L = load resistance (Ω); replace R_2 by this combination in the formula
Current in the divider	$I = \frac{V_{in}}{R_1 + R_2}$; $V = IR$	I = current (A); V = p.d. across each resistor (V), using its own R

TIPS TO ANSWER

- Identify the resistor across which the output is taken, then write Vout = Vin × $\frac{R}{R_1 + R_2}$ and substitute.
- For sensor questions, state the change of resistance, then the change of its fraction of the total resistance, then the change of the output p.d.
- For loading, combine the load and output resistor in parallel first, then recalculate the ratio.
- Give the voltage in V to 3 s.f.; resistances in kΩ may be used directly if they are consistent, as the ratio is dimensionless.

SECTION A TRAPS

Trap: Using the wrong resistor in the numerator: the output across R_2 is Vin $\frac{R_2}{R_1 + R_2}$, not Vin $\frac{R_1}{R_1 + R_2}$.

Trap: Believing that the output voltage remains the same when a load is connected; the parallel load decreases it.

Trap: Believing that an LDR in bright light has an increased resistance; its resistance decreases, so the p.d. across it decreases.

Trap: Using $\frac{Vin}{R_1 + R_2}$ without multiplying by R, giving a current instead of a voltage.

TRICKS

Trick: Identical resistors in series share the supply equally: N equal resistors give $\frac{Vin}{N}$ across each.

Trick: Sensor in the lower position: when its resistance falls, the output falls; swapping the positions reverses the effect.

16.4 Potentiometer and Wheatstone Bridge

DEFINITIONS & LAWS

- **A potentiometer** is a device used to measure the e.m.f. of a cell, or to compare e.m.f.s, by balancing the p.d. across a length of uniform wire against the e.m.f. of the cell, so that no current is drawn from the cell at balance.
- **At the balance point** the galvanometer reads zero, so there is no current in the galvanometer branch and no p.d. across it.
- **The potentiometer** is more accurate than a voltmeter because at balance no current is drawn from the cell, so the measured value equals the e.m.f. and not the terminal p.d.
- **A Wheatstone bridge** is balanced when the galvanometer reads zero, which occurs when the ratio of the resistances in one arm equals the ratio in the other, $\frac{P}{Q} = \frac{R}{S}$.
- **At balance** the two points connected to the galvanometer are at the same potential.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Potentiometer balance length	$E \propto l;\ \frac{E_1}{E_2} = \frac{l_1}{l_2}$	l = balance length (cm or m); for a uniform wire with a constant driver current
E.m.f. from balance length	$E = \frac{l}{L} V_0$	L = total wire length (same unit as l); V_0 = p.d. across the whole wire (V)
P.d. across the wire	$V_0 = IR_{wire}$	I = driver current (A); R_{wire} = resistance of the wire (Ω)
Driver circuit current	$I = \frac{E_{driver}}{R + R_{wire} + r}$	E_{driver} = e.m.f. of the driver cell (V); R = series resistance (Ω); r = internal resistance (Ω)
Wheatstone bridge balance	$\frac{P}{Q} = \frac{R}{S}$	P, Q, R, S = resistances of the four arms (Ω); at balance

Quantity / law	Formula	Symbols, units & conditions
Slide-wire bridge	$\frac{R}{X} = \frac{l_1}{l_2}$	X = unknown resistance (Ω); l_1, l_2 = lengths of the wire on either side of the balance point; resistance is proportional to length for a uniform wire

TIPS TO ANSWER

- State that E is proportional to the balance length because the wire is uniform and carries a constant current.
- Write $\frac{E_1}{E_2} = \frac{l_1}{l_2}$, substitute the lengths and then calculate the unknown e.m.f. with unit.
- Explain zero deflection by stating that there is no current through the galvanometer, hence no current drawn from the cell.
- For the Wheatstone bridge, write the balance condition first, identify the correct arms, then solve for the unknown resistance.
- Driver cell only across the wire: use $V \propto l$ directly. Driver cell with a resistance in series (or internal resistance): first find the current in the driver circuit, $I = E/(R_{total})$, then the p.d. per unit length, then the balance length.
- Meter bridge: keep the balance length between 30 cm and 70 cm, and interchange the two resistors and repeat to cancel end errors. A Wheatstone bridge works only when the ratio of resistances on either side of the galvanometer is equal.

SECTION A TRAPS

Trap: Inverting the ratio: the cell with the longer balance length has the larger e.m.f., so $\frac{E_1}{E_2} = \frac{l_1}{l_2}$, not $\frac{l_2}{l_1}$.

Trap: Writing $P + Q = R + S$, $P \times Q = R \times S$ or $P - Q = R - S$ for the bridge condition; the correct condition is a ratio.

Trap: Using the wrong arm: on a slide wire the unknown resistor is in the ratio with the length on its own side, and its position determines whether R is smaller or larger than the standard.

Trap: Omitting the series resistor when calculating the driver current, so that the p.d. across the slide wire is wrongly taken as the driver e.m.f.

TRICKS

Trick: Sanity check: the unknown cell with the shorter balance length must have the smaller e.m.f.; eliminate options that contradict this.

Trick: In a balanced bridge cross-multiply: P × S = Q × R, which is the same as $\frac{P}{Q} = \frac{R}{S}$; the products of the opposite arms are equal.

Chapter 17

MAGNETIC FIELDS

17.1 Concept of a Magnetic Field

DEFINITIONS & LAWS

- **Magnetic field** is a region in which a moving charge, a current-carrying conductor or a magnetic material experiences a magnetic force.
- **Magnetic field lines** are lines whose tangent at any point gives the direction of the magnetic field (flux density) at that point; outside a magnet they are directed from the north pole to the south pole.
- **Magnetic flux density, B,** is the force acting per unit length on a straight conductor carrying a current of 1 A that is placed perpendicular to the magnetic field.
- **One tesla** is the magnetic flux density that produces a force of 1 N on 1 m of straight conductor carrying a current of 1 A perpendicular to the field.
- For a moving charge: B is the force per unit charge per unit velocity on a charge moving perpendicular to the field, $B = \frac{F}{qv \sin \theta}$.
- **Relative permeability** μ_r of a medium is the ratio of the permeability of the medium to the permeability of free space, $\mu_r = \frac{\mu}{\mu_0}$.
- Magnetic field lines always form closed loops (no magnetic monopoles); they never cross; the closer the lines, the stronger the field.
- A magnetic field is produced by a moving charge (a current) or by a permanent magnet; a stationary charge produces only an electric field.
- **Right-hand grip rule (straight wire):** grip the wire with the right hand, thumb pointing along the conventional current; the curled fingers give the direction of the field lines (concentric circles around the wire).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Magnetic flux density	$B = \frac{F}{IL \sin\theta}$	B = magnetic flux density (T); F = force (N); I = current (A); L = length of conductor in the field (m); θ = angle between the conductor and B
Tesla	$1\ T = 1\ N\ A^{-1}\ m^{-1} = 1\ Wb\ m^{-2}$	T = tesla, the unit of magnetic flux density; Wb = weber
Magnetic field of a straight wire	$B = \frac{\mu_0 I}{2\pi r}$	I = current (A); r = perpendicular distance from the wire (m); $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} (T m A^{-1})

TIPS TO ANSWER

- Define B by quoting the force on a conductor: "force per unit current per unit length, conductor perpendicular to the field".
- When asked to sketch a field pattern: draw closed lines with arrows, closer together where the field is stronger, and label N and S poles or the current direction.
- For a wire, state "concentric circles in a plane perpendicular to the wire, getting further apart as r increases".
- State the direction rule used (right-hand grip) and the direction of the current in the same answer.

SECTION A TRAPS

Trap: Field lines of a bar magnet "start at N and end at S" only outside the magnet; inside the magnet they continue from S to N, so the lines form closed loops.

Trap: A stationary charge has no magnetic field; a charge moving at constant velocity does.

Trap: Magnetic field lines are not the path of a charged particle and not the direction of the force; the force on a moving charge is perpendicular to B.

TRICKS

Trick: For a straight wire $B \propto \frac{I}{r}$: doubling I and doubling r together leaves B unchanged; compare ratios instead of calculating.

Trick: Unit check: T = N A^{-1} m^{-1}; any option giving N A m^{-1} or N A m is wrong.

17.2 Force on a Moving Charge

DEFINITIONS & LAWS

- **Magnetic force on a moving charge:** a charge q moving with velocity v in a magnetic field B experiences a force of magnitude F = qvB sinθ, which is perpendicular to both v and B.
- The magnetic force is always perpendicular to the velocity, so it does no work and does not change the speed or kinetic energy of the particle; it changes only the direction of motion.
- When v is perpendicular to B, the magnetic force provides the centripetal force and the particle moves in a circle; when v is parallel (or antiparallel) to B, the force is zero and the motion is a straight line; at other angles the path is a helix.

- **Fleming's left-hand rule:** first finger = direction of B, second finger = direction of conventional current (direction of motion of a positive charge), thumb = direction of force. For a negative charge the second finger points opposite to its velocity.
- Positive and negative charges of the same velocity are deflected in opposite directions.
- **Velocity selector:** in perpendicular electric and magnetic fields, a particle passes undeflected when qE = qvB, so only particles of speed $v = \frac{E}{B}$ emerge.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Magnetic force on a moving charge	$F = qvB\sin\theta$	F = force (N); q = charge (C); v = speed (m s^{-1}); B = magnetic flux density (T); θ = angle between v and B
Force when v ⊥ B	$F = qvB$	Special case θ = 90°, i.e. v perpendicular to B
Circular motion in a magnetic field	$qvB = \frac{mv^2}{r}$	m = mass (kg); r = radius of the circular path (m)
Radius of circular path	$r = \frac{mv}{qB} = \frac{p}{qB}$	p = momentum (kg m s^{-1}); from $qvB = \frac{mv^2}{r}$
Period of circular motion	$T = \frac{2\pi m}{qB}$	T = period (s); independent of v and r
Frequency of circular motion	$f = \frac{qB}{2\pi m}$	f = frequency (Hz); independent of v and r
Velocity selector	$v = \frac{E}{B}$	E = electric field strength (V m^{-1}); B = flux density (T); v = speed (m s^{-1})
Energy gained from a p.d.	$qV = \frac{1}{2}mv^2$	V = accelerating potential difference (V); gain in kinetic energy equals work done by the p.d.
Speed after acceleration by a p.d.	$v = \sqrt{\frac{2qV}{m}}$	v = speed (m s^{-1}); q = charge (C); V = p.d. (V); m = mass (kg)
Constants for electron	$e = 1.60\times10^{-19}$ C; $m_e = 9.11\times10^{-31}$ kg	e = elementary charge (C); m_e = electron mass (kg)

TIPS TO ANSWER

- Calculation: state $F = qvB\sin\theta$, substitute with SI units, give the magnitude with unit and the direction (Fleming's left-hand rule) where asked.
- For circular motion write "magnetic force = centripetal force, $qvB = \frac{mv^2}{r}$" before deriving $r = \frac{mv}{qB}$.

- State "the force is perpendicular to the velocity, hence no work is done and the speed is constant".
- In a ratio question, write $r \propto \frac{m}{q}$ (same v, B) or $r \propto v/\frac{q}{m}$ and substitute the ratios directly.
- Compare: in an electric field (F = qE along the field) a charge entering perpendicularly follows a parabola and its speed changes; in a magnetic field (F = qvB ⊥ v) it follows a circle and its speed is constant.

SECTION A TRAPS

Trap: Using a negative charge in Fleming's rule without reversing the direction: the second finger points along the motion of a POSITIVE charge, so an electron's force is opposite to that of a proton moving the same way.

Trap: Alpha particle: charge +2e and mass ≈ $4m_p$, so $r_{_}\frac{\alpha}{r_p} = \frac{4}{2} = 2$, not 4 or ½.

Trap: Velocity parallel to B gives zero force, not qvB; confusing sinθ with cosθ (angle taken to the normal).

Trap: Period of circular motion does not depend on speed: doubling v doubles r but T is unchanged.

TRICKS

Trick: $r \propto \frac{mv}{qB}$: build a quick ratio table of m, v, q and B before computing anything.

Trick: Same p.d. accelerates different particles: $v \propto \sqrt{\frac{q}{m}}$; then $r \propto \sqrt{\frac{m}{q}}$ if B is the same.

17.3 Force on a Current-Carrying Conductor

DEFINITIONS & LAWS

- A straight conductor of length L carrying a current I in a magnetic field B experiences a force F = BIL sinθ, perpendicular to both the conductor and the field.
- **Fleming's left-hand rule:** first finger = field B, second finger = conventional current I, thumb = force F (motion); the three are mutually perpendicular.
- The force is a maximum (BIL) when the conductor is perpendicular to B and zero when the conductor is parallel to B.
- The force arises because the field exerts a force on each moving charge carrier; the sum over all the carriers in the length L gives the force on the conductor.

- A rectangular coil of N turns in a uniform field experiences no net force but a torque $\tau = NIAB \sin\varphi$, where φ is the angle between the normal to the coil and B (equivalently $NIAB \cos\alpha$ where α is the angle between the plane of the coil and B).
- The torque is a maximum when the plane of the coil is parallel to B and zero when the plane is perpendicular to B; this is the principle of the moving-coil meter and the d.c. motor.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Force on a current-carrying wire	$F = BIL \sin\theta$	F = force (N); B = flux density (T); I = current (A); L = length of wire in the field (m); θ = angle between the wire and B
Torque on a coil	$\tau = NBIA \sin\varphi$	τ = torque (N m); N = number of turns; A = area of the coil (m^2); φ = angle between the normal to the coil and B
Wire balanced against its weight	$BIL = mg$	m = mass of wire (kg); g = acceleration of free fall ($m\ s^{-2}$); B perpendicular to the wire

TIPS TO ANSWER

- State the formula, substitute in SI units (cm → m, g → kg) and quote the force in newtons with the direction from Fleming's left-hand rule.
- Direction: name the three directions used, e.g. "B into page, I to the right, so force is up the page".
- For a coil: state "net force = 0, but the two equal and opposite forces on opposite sides form a couple, giving a torque".
- For a balancing question equate magnetic force and weight, then solve for B or I.

SECTION A TRAPS

Trap: Using L as the total length of wire instead of the length lying within the field.

Trap: Using the angle between the wire and B correctly: at 30°, $\sin\theta = 0.5$; do not use cos 30°.

Trap: Mixing up the plane and the normal of the coil: torque is maximum when the plane of the coil is PARALLEL to B (normal ⊥ B).

Trap: Reversing the current or field reverses the force; reversing both leaves it unchanged.

TRICKS

> **Trick:** Sketch axes x, y, z: if I and B are along two axes, the force lies along the third; the sign follows from I × B.

> **Trick:** Limiting-case check: θ = 0° gives F = 0, so any option showing a non-zero force for a wire parallel to B is wrong.

17.4 Magnetic Fields Due to Currents

DEFINITIONS & LAWS

- **Ampère's law**: for any closed path, the sum of the products of the length elements Δl and the component of B parallel to them equals μ_0 times the current enclosed by the path, $\Sigma B_{\parallel} \Delta l = \mu_0 I$.
- **Right-hand grip rule (straight wire):** thumb along the conventional current, fingers curl in the direction of B.
- **Right-hand grip rule (coil/solenoid):** fingers curl in the direction of the current, thumb points along B inside the coil, i.e. towards the north-pole end.
- The field of a long straight wire consists of concentric circles; B is inversely proportional to the perpendicular distance r.
- The field inside a long solenoid (far from the ends) is uniform and parallel to the axis; the field outside is very small.
- Superposition: the resultant field at a point is the vector sum of the fields produced by each current; fields from different wires may cancel at a neutral point.
- Two coaxial coils or concentric coils with opposing currents give zero net field at the centre when their individual fields are equal in magnitude.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Field of a long straight wire	$B = \frac{\mu_0 I}{2\pi r}$	B = flux density (T); I = current (A); r = perpendicular distance from the wire (m)
Field at centre of a circular coil	$B = \frac{\mu_0 N I}{2r}$	N = number of turns; r = radius of the coil (m); I = current (A)
Field inside a long solenoid	$B = \mu_0 n I = \frac{\mu_0 N I}{l}$	n = number of turns per metre (m^{-1}); N = total number of turns; l = length of solenoid (m)
Field of a toroid	$B = \frac{\mu_0 N I}{2\pi r}$	r = mean radius of the ring (m); N = number of turns; I = current (A)
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} = 1.26×10^{-6} H m^{-1}	μ_0 = permeability of free space (H m^{-1})

TIPS TO ANSWER

- Quote the formula, substitute with SI units (cm → m) and give B in tesla to 3 s.f., with direction by the right-hand grip rule when asked.
- For a resultant field, find the magnitude and direction of each contribution separately, then add as vectors.
- For graphs of B against r (straight wire) sketch a curve of the form $\frac{1}{r}$: steep at small r, never reaching zero.
- For a solenoid always use $n = \frac{N}{l}$ (turns per metre), not N.
- Neutral point between two parallel wires with currents in the same direction: set $\frac{\mu_0 I_1}{2\pi r} = \mu_0 I_2/(2\pi(d - r))$ and solve for r. Example: 4 A and 3 A, so $\frac{4}{r} = \frac{3}{d - r}$ and $r = \frac{4d}{7}$.

SECTION A TRAPS

Trap: Using the total number of turns N in $B = \mu_0 nI$ instead of turns per metre.

Trap: Writing B for a straight wire $\propto \frac{1}{r^2}$ or $\propto r$; the correct relationship is $B \propto \frac{1}{r}$.

Trap: Rewinding the same length of wire into N turns: radius becomes $\frac{r_0}{N}$, so $B = \frac{\mu_0 NI}{2}\frac{r_0}{N}$ increases by N^2 (not N).

Trap: Forgetting the 2π (wire) or the 2 (coil) in the denominator; the solenoid formula has neither.

TRICKS

Trick: Midway between two wires with currents in the same direction the fields cancel; for opposite directions they add.

Trick: Same-length rewind: $B \propto N^2$; doubling the number of turns per metre of a solenoid doubles B.

17.5 Force Between Two Current-Carrying Conductors

DEFINITIONS & LAWS

- The field of one wire exerts a force on the current in the other; by Newton's third law the two forces are equal in magnitude and opposite in direction, whatever the sizes of the currents.

- Parallel wires carrying currents in the same direction attract each other; wires carrying currents in opposite directions repel each other.
- **The ampere** is that constant current which, if maintained in two straight parallel conductors of infinite length and negligible cross-section placed one metre apart in a vacuum, produces a force of 2×10^{-7} N per metre of length between them.
- Explanation: wire 1 produces a field $B_1 = \frac{\mu_0 I_1}{2\pi d}$ at wire 2; the force on wire 2 is $F = B_1 I_2 L$, directed by Fleming's left-hand rule towards wire 1 when the currents are parallel.
- **The tesla** is the magnetic flux density that gives a force of 1 N per metre on a conductor carrying 1 A perpendicular to the field.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Force per unit length between wires	$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$	$\frac{F}{L}$ = force per unit length (N m^{-1}); I_1 and I_2 = currents (A); d = separation (m)
Definition check (ampere)	$\frac{F}{L} = \frac{4\pi\times10^{-7}}{2\pi} = 2\times10^{-7}$ N m^{-1}	For $I_1 = I_2 = 1$ A and d = 1 m
Force on a length of wire	$F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$	l = length of one wire (m); I_1 and I_2 = currents (A); d = separation (m)

TIPS TO ANSWER

- Show the field of one wire at the position of the other, then F = BIL, then direction by Fleming's left-hand rule.
- State the nature: "attractive" for currents in the same direction, "repulsive" for opposite directions, with the magnitude per metre.
- In the definition of the ampere include all four conditions: constant current, infinite length and negligible cross-section, 1 m apart, in a vacuum; the force is 2×10^{-7} N per metre.
- Substitute with SI units and give $\frac{F}{L}$ in N m^{-1} to 3 s.f.; if a length l is given, multiply by l.
- Rectangular loop beside a long wire: the forces on the two sides perpendicular to the wire cancel; the side nearer the wire is in a stronger field ($B = \frac{\mu_0 I}{2\pi r}$), so the net force is towards the wire when the nearer side carries current in the same direction as the wire.

SECTION A TRAPS

Trap: Same-direction currents REPEL: this is wrong; parallel currents attract (opposite currents repel).

Trap: Claiming the wire with the larger current feels the larger force; the two forces are equal and opposite.

Trap: Ratio questions: $\frac{F}{L} \propto \frac{I_1 I_2}{d}$, so doubling both currents and d gives $\frac{4}{2}$ = 2 times, not 4 or 1.

Trap: Using $\frac{\mu_0}{2\pi d}$ with d in cm; convert to metres first.

TRICKS

Trick: Use the ratio method: write $F \propto \frac{I_1 I_2}{d}$ and substitute the multipliers.

Trick: Since $\frac{\mu_0}{2\pi} = 2\times10^{-7}$ T m A^{-1}, $\frac{F}{L} = 2\times10^{-7} \frac{I_1 I_2}{d}$ with d in metres; this avoids handling 4π.

17.6 Determination of the Ratio e/m

DEFINITIONS & LAWS

- **Specific charge** of the electron is the ratio of its charge to its mass, $\frac{e}{m}$ (C kg^{-1}).
- Thomson's method: electrons are accelerated and pass through crossed electric and magnetic fields; the fields are adjusted so that the beam is undeflected, so that eE = evB, giving $v = \frac{E}{B}$.
- With the electric field switched off, the magnetic force makes the electrons move in a circular arc: $evB = \frac{mv^2}{r}$, so $\frac{e}{m} = v/(Br)$.
- Alternative method: electrons accelerated through p.d. V gain kinetic energy $eV = \frac{1}{2}mv^2$; combined with $r = \frac{mv}{eB}$ this gives $\frac{e}{m} = 2V/(B^2r^2)$.
- The experiment determines the ratio $\frac{e}{m}$ only; the charge e must be found independently (Millikan's oil-drop experiment) to find m.
- The ratio $\frac{e}{m}$ for cathode rays is the same for all cathode materials, showing that electrons are a universal constituent of matter.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Balanced fields (velocity)	$eE = evB;\ v = \frac{E}{B}$	e = electron charge (C); E = electric field strength (V m^{-1}); B = flux density (T); v = speed (m s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Magnetic deflection	$r = \frac{mv}{eB}$; $\frac{e}{m} = \frac{v}{rB}$	r = radius of circular path (m); m = electron mass (kg)
Accelerating p.d.	$eV = \frac{1}{2}mv^2$; $v = \sqrt{\frac{2eV}{m}}$	V = accelerating p.d. (V)
Combined expression for $\frac{e}{m}$	$\frac{e}{m} = 2V/(B^2r^2)$	$\frac{e}{m}$ = specific charge of the electron (C kg^{-1})
Accepted value of $\frac{e}{m}$	$\frac{e}{m} = 1.76\times10^{11}$ C kg^{-1}	Accepted value of the specific charge (C kg^{-1})

TIPS TO ANSWER

- Explain the first step: "balance electric and magnetic forces so that the beam is undeflected, hence v = $\frac{E}{B}$"; this gives the speed independently of $\frac{e}{m}$.
- Describe the second step: "remove the electric field; the beam moves in a circle of radius r; magnetic force provides centripetal force".
- For calculations write $\frac{e}{m} = 2V/(B^2r^2)$, substitute with SI units and give the answer to 3 s.f. in C kg^{-1} with the order of magnitude 10^{11}.
- For error analysis: $\frac{e}{m} \propto V/(B^2r^2)$, so a fractional error in B or r counts twice.
- A short question ("derive $\frac{e}{m}$") needs a complete answer: name the two parts (speed selector and deflection chamber), state the condition used in each ($eE = evB$, or $eV = \frac{1}{2}mv^2$; then $evB = \frac{mv^2}{r}$), and combine.

SECTION A TRAPS

Trap: Forgetting that r and B are squared in $\frac{e}{m} = 2V/(B^2r^2)$; the squares also double fractional errors.

Trap: Giving the answer in the wrong power of ten: a result such as 10^{10} or 10^{12} C kg^{-1} signals an arithmetic slip; the accepted value is 1.76×10^{11} C kg^{-1}.

Trap: Claiming that $\frac{e}{m}$ experiments measure e or m alone; only the ratio is found.

Trap: Radius vs diameter: if the diameter of the circular path is measured, halve it before substituting.

TRICKS

Trick: The $\frac{e}{m}$ value must be near 10^{11}: check the order of magnitude after calculating.

Trick: V is directly proportional to $\frac{e}{m}$: a reading of V too high by 10% gives $\frac{e}{m}$ too high by 10%.

17.7 Hall Effect

DEFINITIONS & LAWS

- **Hall effect** is the production of a potential difference (Hall voltage) across a current-carrying conductor or semiconductor placed in a magnetic field, the potential difference being perpendicular to both the current and the field.
- **Hall voltage** is the transverse potential difference produced across a current-carrying conductor when a magnetic field is applied perpendicular to the current, as the drifting charge carriers are deflected by the magnetic force.
- Explanation: the magnetic force deflects the moving charge carriers to one side of the slab; the accumulated charge sets up a transverse electric field E_H.
- The Hall voltage reaches a steady value when the electric force on a carrier balances the magnetic force: $eE_H = evB$.
- The sign of the Hall voltage identifies the sign of the charge carriers: with the same current direction and field direction, electrons and holes are deflected to the same side but give opposite polarity across the slab (use Fleming's left-hand rule with conventional current).
- In a semiconductor the carrier density n is much lower than in a metal, so V_H is much larger.
- Application: a Hall probe measures B because $V_H \propto B$ at constant current; it is made from a thin semiconductor slab for a large V_H.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Hall voltage	$V_H = \frac{BI}{nte}$	V_H = Hall voltage (V); B = flux density (T); I = current (A); n = number density of carriers (m^{-3}); t = thickness of the slab in the direction of B (m); e = 1.60×10^{-19} C
Balance of forces	$\frac{eV_H}{d} = evB$	d = width of the slab between the Hall faces (m); v = drift velocity ($m\ s^{-1}$)
Current in terms of drift velocity	$I = nevtd$	Substituting into the force balance gives $V_H = \frac{BI}{nte}$

Quantity / law	Formula	Symbols, units & conditions
Drift velocity	$v = \frac{I}{nAe}$	A = cross-sectional area (m^2); n = number density of carriers (m^{-3}); e = 1.60×10^{-19} C

TIPS TO ANSWER

- Explain by steps: carriers move, magnetic force deflects them to one face, charge accumulates, transverse E field builds up until $eE_H = evB$.
- State the polarity: for electrons deflected by the field, the face where they accumulate is negative; for holes the opposite face becomes positive; mention the direction rule used.
- For calculations write $V_H = \frac{BI}{nte}$, substitute with SI units (mm → m) and give V_H in volts (μV or mV) to 3 s.f.
- Explain why a semiconductor is used: low n gives large V_H (and a thin slab, small t, also raises V_H).
- Decide the carriers from the words: a metal (conductor) → electrons; a p-type semiconductor → positive holes. Apply Fleming's left-hand rule to the carriers and label the positive and negative faces on a diagram; the polarity marks need the diagram even if you use $V_H = \frac{BI}{nqt}$.

SECTION A TRAPS

Trap: t is the thickness in the direction of B, not the width of the strip; using the wrong dimension gives the wrong V_H.

Trap: Taking $V_H \propto n$: V_H is inversely proportional to n, so a low density of carriers gives a LARGER Hall voltage.

Trap: Forgetting to convert mm to m (0.10 mm = 1.0×10^{-4} m); metals give V_H of order μV.

Trap: The Hall voltage does not depend on the length of the slab or the sign of the carriers in magnitude; only the polarity depends on the sign.

TRICKS

Trick: $V_H \propto \frac{BI}{nt}$: doubling I doubles V_H; doubling t halves V_H.

Trick: Estimate order of magnitude: for metals $n \approx 10^{29}$ m^{-3} gives $V_H \approx$ μV; semiconductors ($n \approx 10^{20}$ m^{-3}) give mV.

Chapter 18

ELECTROMAGNETIC INDUCTION

18.1 Magnetic Flux

DEFINITIONS & LAWS

- **Magnetic flux** through a surface is the product of the magnetic flux density and the area of the surface, measured perpendicular to the field, $\Phi = BA\cos\theta$; it can be pictured as the total number of magnetic field lines passing through the area.
- **Weber (Wb)** is the magnetic flux when a magnetic flux density of 1 T passes perpendicularly through an area of 1 m^2.
- **Magnetic flux linkage** is the product of the number of turns of a coil and the magnetic flux through each turn, $N\Phi$.
- **Magnetic flux density** B is the magnetic flux per unit area perpendicular to the field (the number of field lines per unit normal area), unit tesla ($T = Wb\ m^{-2}$).
- For a non-uniform field, the flux is found by summing the normal component of B over all area elements, $\Phi = \int B \cdot dA$.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Magnetic flux	$\Phi = BA\cos\theta$	Φ = magnetic flux (Wb = T m^2); B = magnetic flux density (T); A = area (m^2); θ = angle between B and the normal to the plane
Flux linkage	$N\Phi = NBA\cos\theta$	$N\Phi$ = flux linkage (Wb-turns); N = number of turns
Maximum flux	$\Phi = BA$ at $\theta = 0°$	Plane perpendicular to B
Zero flux	$\Phi = 0$ at $\theta = 90°$	Plane parallel to B
Area of circular coil	$A = \pi r^2$	A = area (m^2); r = radius (m)
Area of square coil	$A = (\text{side})^2$	A = area (m^2); side in m

TIPS TO ANSWER

- State the formula $\Phi = BA\cos\theta$, define θ as the angle between B and the normal to the coil, then substitute with SI units and give the answer in Wb.
- For flux linkage multiply by N and state the unit as Wb turns (or Wb).

- Convert cm^2 to m^2 ($1\ cm^2 = 10^{-4}\ m^2$) and radius to metres before calculating the area.
- In a descriptive answer, state that the flux depends on the field strength, the area and the orientation of the coil relative to the field.

SECTION A TRAPS

Trap: The angle θ is taken between B and the plane of the coil. This is wrong: θ is between B and the normal. A plane parallel to B has zero flux, not maximum flux.

Trap: Using $A = \pi d^2$ with the diameter, or leaving the area in cm^2, gives an answer wrong by a factor of 4 or 10^4.

Trap: Forgetting N when the question asks for flux linkage; the flux through one turn is not the flux linkage.

TRICKS

Trick: Plane perpendicular to B gives the maximum $\Phi = BA$; plane parallel to B gives $\Phi = 0$. Any $\cos\theta$ value between these can be checked as a fraction of BA ($\cos 60° = 0.5$, $\cos 30° = 0.866$).

18.2 Faraday's Law and Lenz's Law

DEFINITIONS & LAWS

- **Faraday's law**: the magnitude of the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage (or the rate of cutting of magnetic flux).
- **Electromagnetic induction** is the production of an induced e.m.f. (and an induced current in a closed circuit) when the magnetic flux linked with a conductor changes.
- **Induced e.m.f.** is the e.m.f. produced in a conductor when there is a change in the magnetic flux linked with it.
- **Lenz's law**: the direction of the induced current is such that it opposes the change in magnetic flux that produces it.
- Lenz's law is a consequence of the conservation of energy: work must be done against the opposing effect to produce the induced current.
- Induced e.m.f. appears whenever the flux linkage changes, by changing B, A or θ, whether or not the circuit is closed; an induced current flows only in a closed circuit.
- For a straight conductor moving across a uniform field, the induced e.m.f. is the product of B, the length of the conductor and the velocity perpendicular to B.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Faraday's law	$\varepsilon = -N\frac{d\Phi}{dt} = -\frac{d(N\Phi)}{dt}$	ε = induced emf (V); N = number of turns; Φ = magnetic flux (Wb); t = time (s); the minus sign represents Lenz's law
Faraday's law, uniform change	$\lvert\varepsilon\rvert = \frac{N\Delta\Phi}{\Delta t}$	ε = induced emf (V); N = number of turns; ΔΦ = change in flux (Wb); Δt = time interval (s); for a uniform change
Emf induced in a moving rod	$\varepsilon = BLv$	ε = emf (V); B = magnetic flux density (T); L = length of rod (m); v = velocity of rod (m s^{-1}); B, L and v mutually perpendicular
Emf with v at angle to B	$\varepsilon = BLv\sin\theta$	θ = angle between v and B; other symbols as above
Rod in a closed circuit	$I = \frac{BLv}{R}$; $F = BIL = \frac{B^2L^2v}{R}$; $P = Fv = \frac{\varepsilon^2}{R}$	I = current (A); R = resistance of circuit (Ω); F = force (N); P = power (W); B, L, v as above; ε = BLv
Emf in a rotating coil	$\varepsilon = NBA\omega\sin\omega t$; $\varepsilon_0 = NBA\omega$	ε = induced emf (V); ε_0 = peak emf (V); N = number of turns; B = magnetic flux density (T); A = area (m^2); ω = angular speed (rad s^{-1}); uniform field

TIPS TO ANSWER

- For a calculation, state $\varepsilon = -N\frac{d\Phi}{dt}$, evaluate ΔΦ (final minus initial, with area and angle), divide by Δt, and give the magnitude in V with unit.
- For direction, apply Lenz's law in three steps: state the change in flux (increase or decrease), state that the induced field opposes this change, then deduce the current direction (right-hand grip rule) or the pole formed.
- Pole rule: an approaching pole induces the same pole at the near end of the coil (repulsion); a receding pole induces the opposite pole (attraction).
- In 'explain why' answers, mention the conservation of energy: if the induced current assisted the change, energy would be created.
- A copper plate falling past a strong magnet: the changing flux induces eddy currents whose magnetic force opposes the motion (Lenz's law), so its acceleration is less than g.
- When asked for the external force needed to keep a rod moving at constant speed, give its value: $P = \frac{\varepsilon^2}{R} = Fv$, so $F = B^2I^2v/R$ (not just "push" or "pull").

SECTION A TRAPS

Trap: Writing that the induced e.m.f. is proportional to the magnetic flux. It is proportional to the rate of change of flux; a large but constant flux gives zero e.m.f.

Trap: Stating that the induced current opposes the magnetic field or the flux. It opposes the change in flux; when the flux decreases, the induced field is in the same direction as the original field.

Trap: Forgetting the factor N for a coil, or using the final flux instead of the change in flux, $\Delta\Phi$.

Trap: Doubling v in a rod and rails circuit doubles ε and I, but the power $\frac{\varepsilon^2}{R}$ and the force on the rod change by different factors: P becomes 4 times, F becomes 2 times.

TRICKS

Trick: Because $P = \frac{\varepsilon^2}{R}$ and $\varepsilon \propto v$, any change in speed by a factor k changes the power by k^2, while the retarding force changes by k. Check by ratio instead of recalculating.

18.3 Self Induction

DEFINITIONS & LAWS

- **Self-induction** is the production of an induced e.m.f. in a coil due to a change in the current in the same coil, which changes the magnetic flux linked with it.
- **Self-inductance** L of a coil is the ratio of the magnetic flux linkage in the coil to the current flowing through it, $L = \frac{N\Phi}{I}$.
- **Henry (H)** is the self-inductance of a coil in which an e.m.f. of 1 V is induced when the current changes at the rate of 1 A s^{-1}.
- Self-inductance can also be defined as the back e.m.f. induced in a coil per unit rate of change of current in the coil, $L = -\varepsilon/\frac{dI}{dt}$.
- **Back e.m.f.** is the induced e.m.f. produced in a circuit by a change in its own current; it opposes the change in current that produces it (Lenz's law).
- By Lenz's law, the self-induced e.m.f. always opposes the change in current: it opposes an increase in current and opposes a decrease in current.
- The self-induced e.m.f. is large when the current is switched off very quickly, since $\frac{dI}{dt}$ is large; this explains the spark across the switch contacts.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Self-induced emf	$\varepsilon = -L\frac{dI}{dt}$	ε = induced emf (V); L = self-inductance (H); $\frac{dI}{dt}$ = rate of change of current (A s^{-1})

Quantity / law	Formula	Symbols, units & conditions
Self-induced emf, uniform change	$\lvert\varepsilon\rvert = L\,\frac{\Delta I}{\Delta t}$	ΔI = change in current (A); Δt = time interval (s); for a uniform change
Self-inductance	$L = \frac{N\Phi}{I}$	L = self-inductance (H = Wb A^{-1} = V s A^{-1}); N = number of turns; Φ = flux (Wb); I = current (A)
Inductance of air-cored solenoid	$L = \frac{\mu_0 N^2 A}{l}$	L = self-inductance (H); N = number of turns; A = cross-sectional area (m^2); l = length (m); μ_0 = $4\pi\times10^{-7}$ H m^{-1}; long air-cored solenoid
Inductance with a core	$L = \frac{\mu_r \mu_0 N^2 A}{l}$	μ_r = relative permeability of the core (no unit); other symbols as above

TIPS TO ANSWER

- Define L using flux linkage and current $\frac{N\Phi}{I}$, not e.m.f. and current alone; you may add that $\varepsilon = -L\,\frac{dI}{dt}$.
- For a calculation, state $\lvert\varepsilon\rvert = L\,\frac{\Delta I}{\Delta t}$, take ΔI as the magnitude of the change in current (final minus initial), substitute with units and give the answer in V.
- In an explanation of the delay in the growth of current, state: the changing current induces an e.m.f. which opposes the change, so the current rises or falls gradually and not instantly.
- For $L = \frac{\mu_0 N^2 A}{l}$, always state which quantity changes and apply the proportionality: $L \propto N^2$, $L \propto A$, $L \propto \frac{1}{l}$.

SECTION A TRAPS

Trap: Using $L \propto N$ instead of $L \propto N^2$. Doubling the turns on a fixed length gives 4 times the inductance.

Trap: Stretching the solenoid to twice the length with N and A fixed halves L, since $L \propto \frac{1}{l}$; it does not double it.

Trap: Stating that the self-induced e.m.f. opposes the current. It opposes the change in current: it opposes an increase and, when the current decreases, it acts in the same direction as the current to oppose the decrease.

Trap: Taking $\Delta I = 0$ because the final current is zero; ΔI is final minus initial, for example 6.0 − 2.0 = 4.0 A, not 6.0 A.

TRICKS

> **Trick:** Check unit consistency: H = V s A^{-1}, so e.m.f. = H × A s^{-1} gives V automatically; if the answer comes out in mV or kV, check the powers of ten in Δt.

18.4 Energy Stored in an Inductor

DEFINITIONS & LAWS

- The energy stored in an inductor is the work done against the self-induced e.m.f. in establishing the current; it is stored in the magnetic field of the inductor.
- The energy is released when the current decreases, as the self-induced e.m.f. then maintains the current for a short time.
- The energy stored per unit volume of a magnetic field is the energy density, $u = \frac{B^2}{2\mu_0}$.
- Energy is stored in the magnetic field of the inductor, not lost; in an ideal inductor there is no energy dissipation.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Energy stored in an inductor	$E = \frac{1}{2}LI^2$	E = energy (J); L = self-inductance (H); I = current (A)
Energy in terms of flux linkage	$E = \frac{N\Phi I}{2}$	N = number of turns; Φ = flux (Wb); I = current (A); since NΦ = LI
Magnetic energy density	$u = \frac{B^2}{2\mu_0}$	u = energy density (J m^{-3}); B = magnetic flux density (T); $\mu_0 = 4\pi\times10^{-7}$ H m^{-1}
Average power released	$P = \frac{E}{t}$	P = average power (W); E = energy (J); t = time (s)

TIPS TO ANSWER

- State $E = \frac{1}{2}LI^2$, substitute with SI units (L in H, I in A) and give the answer in J with 3 s.f.
- For a current changing from I_1 to I_2, calculate the change in energy as $\frac{1}{2}L(I_2^2 - I_1^2)$; do not square the difference of the currents.
- To find I or L from a given energy, rearrange first: $I = \sqrt{\frac{2E}{L}}$ and $L = \frac{2E}{I^2}$.
- For power over a time interval, state $P = \frac{E}{t}$ with the energy released in that time.

SECTION A TRAPS

Trap: Using $E = LI^2$ or $E = \frac{1}{2}LI$ (missing the factor ½ or the square), or $E = \frac{1}{2}L^2I$.

Trap: Squaring after doubling incorrectly: doubling I multiplies E by 4, and tripling I multiplies E by 9, but doubling L only doubles E.

Trap: In an energy-density question, forgetting the factor 2 in the denominator, or taking $\mu_0 = 4\pi\times10^{-7}$ without the 4π.

TRICKS

Trick: Use proportion: $E \propto L$ at fixed I and $E \propto I^2$ at fixed L. For example, I tripled gives 9 times the energy; compare options by ratio before evaluating.

18.5 Mutual Induction

DEFINITIONS & LAWS

- **Mutual induction** is the production of an induced e.m.f. in one coil due to a change in the current in a neighbouring coil, which changes the magnetic flux linked with the first coil.
- **Mutual inductance** M is the ratio of the magnetic flux linkage in one coil to the current in the other coil that produces the flux, $M = \frac{N_2\Phi_{21}}{I_1}$.
- **Henry (H)** is the mutual inductance of two coils when a current changing at the rate of 1 A s^{-1} in one coil induces an e.m.f. of 1 V in the other.
- Mutual inductance can also be defined as the e.m.f. induced in one coil per unit rate of change of current in the other coil, $M = -\varepsilon_2/\frac{dI_1}{dt}$.
- Mutual inductance depends on the geometry of the coils (number of turns, size, separation, relative orientation) and the permeability of the core, not on the current.
- **Transformer**: an alternating current in the primary coil produces a changing flux in the core, which induces an alternating e.m.f. in the secondary coil by mutual induction.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Mutually induced emf	$\varepsilon_2 = -M\frac{dI_1}{dt}$	ε_2 = emf induced in secondary (V); M = mutual inductance (H); $\frac{dI_1}{dt}$ = rate of change of primary current (A s^{-1})
Mutually induced emf, uniform change	$\lvert\varepsilon_2\rvert = M\frac{\Delta I_1}{\Delta t}$	ΔI_1 = change in primary current (A); Δt = time interval (s); for a uniform change

Quantity / law	Formula	Symbols, units & conditions
Mutual inductance	$N_2\Phi_{21} = MI_1$	N_2 = number of turns of secondary; Φ_{21} = flux in secondary due to primary (Wb); M = mutual inductance (H = Wb A^{-1}); I_1 = primary current (A)
Ideal transformer	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$; $V_pI_p = V_sI_s$	V = voltage (V); N = number of turns; I = current (A); subscripts p = primary, s = secondary
Coupling coefficient	$M = k\sqrt{L_1L_2}$	M = mutual inductance (H); k = coupling coefficient, $0 \le k \le 1$; L_1, L_2 = self-inductances of the two coils (H)

TIPS TO ANSWER

- Define M with the flux linkage in one coil and the current in the other coil (cross-reference of two coils), not the same coil.
- State $\varepsilon = -M\frac{dI}{dt}$, substitute M and the rate of change of current $\frac{\Delta I}{\Delta t}$ with units, and give the answer in V.
- For a transformer, state that the changing flux in the core links both coils, and that $\frac{V_s}{V_p} = \frac{N_s}{N_p}$; identify step-up ($N_s$ greater than N_p) or step-down.
- Explain that a steady direct current in the primary induces no e.m.f. in the secondary because the flux does not change.
- Explain mutual induction in sequence: current in the primary coil produces a magnetic flux; the flux links the secondary coil; when the primary current changes, the flux linkage in the secondary changes; an e.m.f. is induced in the secondary (Faraday's law).

SECTION A TRAPS

Trap: Writing that M depends on the magnitude of the primary current. It does not; only the induced e.m.f. depends on the rate of change of current.

Trap: A transformer ratio inverted: $V_s = V_p \times \frac{N_p}{N_s}$. The secondary voltage is directly proportional to the secondary turns.

Trap: Believing that a constant rate of change of current gives an increasing induced e.m.f. If $\frac{dI}{dt}$ is constant, ε_2 is constant.

Trap: Not converting mH to H (25 mH = 25×10^{-3} H) or ms to s.

TRICKS

> **Trick:** For a transformer, check the sign of the answer by comparing turns: fewer secondary turns means a lower voltage. A 20:1 turns ratio on 240 V gives 12 V.

Chapter 19

ALTERNATING CURRENT CIRCUITS

19.1 AC Through a Pure Resistor

DEFINITIONS & LAWS

- An **alternating current** is a current that reverses its direction at a constant frequency; usually it varies sinusoidally with time, $I = I_0 \sin\omega t$.
- **Root-mean-square (r.m.s.) value** of an alternating current is the value of the steady direct current that dissipates heat in a given resistor at the same rate as the alternating current.
- In a pure resistor, the current is in phase with the voltage.
- The instantaneous power $P = I^2R$ varies with a frequency twice that of the current and is always non-negative; the mean power is half the peak power.
- Peak value: the maximum magnitude of the current or voltage during a cycle.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Current and voltage	$I = I_0 \sin\omega t$; $V = V_0 \sin\omega t$	I = current (A); V = voltage (V); I_0, V_0 = peak values; $\omega = 2\pi f$ (rad s^{-1}); f = frequency (Hz); t = time (s)
Period	$T = \frac{1}{f}$	T = period (s); f = frequency (Hz)
r.m.s. values	$I_{rms} = \frac{I_0}{\sqrt{2}}$; $V_{rms} = \frac{V_0}{\sqrt{2}}$	I_{rms} = r.m.s. current (A); V_{rms} = r.m.s. voltage (V); I_0, V_0 = peak values; sinusoidal waveform only
Peak current from r.m.s.	$I_0 = \sqrt{2}\, I_{rms} \approx 1.41\, I_{rms}$	I_0 = peak current (A); I_{rms} = r.m.s. current (A)
Mean power	$P = I_{rms}^2 R = \frac{V_{rms}^2}{R} = I_{rms} V_{rms}$ $= \frac{1}{2} I_0^2 R = \frac{1}{2} V_0 I_0$	P = mean power (W); R = resistance (Ω); I_{rms}, V_{rms} = r.m.s. current (A) and voltage (V); I_0, V_0 = peak values
Peak power	$P_{peak} = I_0^2 R = 2P_{mean}$	P_{peak} = peak power (W); P_{mean} = mean power (W); I_0 = peak current (A); R = resistance (Ω)
Energy dissipated	$E = P_{mean}\, t$	E = energy (J); P_{mean} = mean power (W); t = time (s)

TIPS TO ANSWER

- State the r.m.s. definition in terms of the heating effect of a steady d.c. in the same resistor.

- From $V = V_0 \sin(\omega t)$ identify V_0 as the coefficient of the sine and ω as the coefficient of t; then $f = \frac{\omega}{2\pi}$ and $V_{rms} = \frac{V_0}{\sqrt{2}}$.
- State $P = \frac{V_{rms}^2}{R}$ using r.m.s. values; if the peak value is given, convert first.
- For a sketch of V and I against t, show that the two curves are in phase, with labelled V_0, I_0 and period T.
- Rate of heat dissipation in a resistor = mean power = $I_{rms}^2R = \frac{1}{2}I_0^2R$ (not I_0^2R).
- Choosing a fuse: heating depends on the r.m.s. current, so choose a fuse rating slightly above I_{rms} (e.g. I_{rms} = 8.7 A → 10 A fuse).
- Write an a.c. equation with numbers, not in terms of π, unless asked: V = 325 sin(314t), not V = 325 sin(100πt).

SECTION A TRAPS

Trap: Using the peak value in $P = \frac{V^2}{R}$ gives the peak power, which is twice the mean power. Convert to r.m.s. first.

Trap: Reading ω (e.g. 314 or 377 rad s^{-1}) as the frequency in Hz. The frequency is $f = \frac{\omega}{2\pi}$ (50 Hz or 60 Hz).

Trap: Taking the mean voltage of a sinusoid as $\frac{V_0}{\sqrt{2}}$; the mean value over a full cycle is zero, and $\frac{V_0}{\sqrt{2}}$ is the r.m.s. value.

Trap: Not converting the time to seconds when calculating energy dissipated, for example 2.0 minutes = 120 s.

TRICKS

Trick: Mean power is always half of the peak power for a resistor: $P_{mean} = \frac{1}{2}P_{peak}$. Use this to eliminate the option that equals the peak and the option that is a quarter of it.

19.2 AC Through a Pure Inductor

DEFINITIONS & LAWS

- **Inductive reactance** X_L is the opposition of an inductor to alternating current, given by the ratio of the r.m.s. (or peak) voltage across the inductor to the r.m.s. (or peak) current through it.

- In a pure inductor, the current lags the applied voltage by 90° ($\frac{\pi}{2}$ rad).
- The mean power dissipated in a pure inductor is zero over one complete cycle, because energy is stored in the magnetic field and returned to the source alternately.
- At very high frequency, the inductor opposes the current strongly; for direct current (f = 0), $X_L = 0$ and the inductor offers no opposition (apart from its resistance).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Inductive reactance	$X_L = \omega L = 2\pi fL$	X_L = inductive reactance (Ω); ω = angular frequency (rad s^{-1}); f = frequency (Hz); L = inductance (H)
Current in an inductor	$I_{rms} = \frac{V_{rms}}{X_L}$; $I_0 = \frac{V_0}{X_L}$	I_{rms}, V_{rms} = r.m.s. current (A) and voltage (V); I_0, V_0 = peak values; X_L = inductive reactance (Ω)
Mean power in a pure inductor	$P = V_{rms} I_{rms} \cos\varphi = 0$	P = mean power (W); φ = phase angle = 90°; V_{rms}, I_{rms} = r.m.s. voltage (V) and current (A)

- X_L is directly proportional to f: a graph of X_L against f is a straight line through the origin with gradient 2πL.

TIPS TO ANSWER

- For a calculation, state $X_L = 2\pi fL$, convert L to henry (mH → $\times 10^{-3}$), substitute and give the answer in Ω; then use $I = \frac{V}{X_L}$ with r.m.s. values.
- State the phase relation exactly: the current lags the voltage by 90° ($\frac{\pi}{2}$ rad), with the correct phrase for the inductor.
- To explain why the mean power is zero: energy is stored in the magnetic field in one quarter-cycle and returned to the source in the next, so the net energy transfer is zero.
- Sketch: sinusoidal V and I, same amplitude scale, with I displaced a quarter-period to the right of V.

SECTION A TRAPS

Trap: Stating that the current leads the voltage in an inductor; that is the capacitor. In an inductor, the voltage leads the current.

Trap: Using $X_L = \frac{1}{\omega L}$, inverted, or forgetting 2π (reading ω as f). Always check that $X_L = 62.8$ Ω for 0.20 H at 50 Hz.

Trap: Using the formula $P = I^2 X_L$ for the power, which suggests heating; the mean power is zero.

Trap: Calculating the current with the peak voltage and then quoting it as r.m.s. Match the peak with peak, r.m.s. with r.m.s.

TRICKS

Trick: ELI the ICE man: in an inductor (L) the e.m.f. (E) leads the current (I); in a capacitor (C) the current leads the e.m.f.

Trick: Doubling f doubles X_L and halves the current at a fixed supply voltage; use ratios to avoid recalculation.

19.3 AC Through a Pure Capacitor

DEFINITIONS & LAWS

- **Capacitive reactance** X_C is the opposition of a capacitor to alternating current, given by the ratio of the r.m.s. (or peak) voltage across the capacitor to the r.m.s. (or peak) current through it.
- In a pure capacitor, the current leads the applied voltage by 90° ($\frac{\pi}{2}$ rad).
- The mean power dissipated in a pure capacitor is zero over one complete cycle, because energy is stored in the electric field and returned to the source alternately.
- A capacitor blocks direct current ($X_C \to \infty$ as $f \to 0$) and passes high-frequency alternating current more easily ($X_C \to 0$ as $f \to \infty$).

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Capacitive reactance	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$	X_C = capacitive reactance (Ω); ω = angular frequency (rad s^{-1}); f = frequency (Hz); C = capacitance (F)
Current in a capacitor (r.m.s.)	$I_{rms} = \frac{V_{rms}}{X_C}$	I_{rms} = r.m.s. current (A); V_{rms} = r.m.s. voltage (V); X_C = capacitive reactance (Ω)
Current in a capacitor (peak)	$I_0 = \frac{V_0}{X_C} = \omega C V_0$	I_0 = peak current (A); V_0 = peak voltage (V); X_C = capacitive reactance (Ω)
Reactance and frequency	$X_C \propto \frac{1}{f}$	X_C is inversely proportional to f; a graph of X_C against f is a decreasing curve (rectangular hyperbola) approaching zero at high frequency
Mean power in a capacitor	$P = V_{rms} I_{rms} \cos\varphi = 0$	P = mean power (W); φ = phase angle between V and I = 90° for a pure capacitor, so cosφ = 0

TIPS TO ANSWER

- For a calculation, state $X_C = \frac{1}{2\pi fC}$, convert C to farad (μF → ×10^{-6}), substitute and give the answer in Ω; then use $I = \frac{V}{X_C}$ with r.m.s. values.
- State the phase relation exactly: the current leads the voltage by 90° ($\frac{\pi}{2}$ rad).
- To explain the variation of current with frequency: as f increases, X_C decreases, so the current increases for a constant r.m.s. voltage.
- On a sketch of X_C against f, label the axes and show the curve falling steeply and approaching, but never touching, the f axis.
- To find C from peak readings: $X_C = V_{rms}/I_{rms}$ (with $V_{rms} = \frac{V_0}{\sqrt{2}}$ and $I_{rms} = \frac{I_0}{\sqrt{2}}$), then $C = 1/(2\pi f X_C)$. Be ready to sketch I and V against t with I leading by $\frac{\pi}{2}$.

SECTION A TRAPS

Trap: Not converting μF to F; 10 μF = 10×10^{-6} F. An answer that is 10^6 times too large or too small indicates this slip.

Trap: Writing $X_C = 2\pi fC$ (inverting); X_C decreases with frequency, but X_L increases.

Trap: Stating that the current in the capacitor increases because the reactance increases; the reactance decreases as f increases.

Trap: Giving the capacitor current at f → 0 as a maximum; $X_C \to \infty$, so the current is zero for steady d.c.

TRICKS

Trick: Frequency doubled: X_L doubles but X_C halves. Both effects are reciprocal; use ratios to check the numerical answer quickly.

19.4 R-C and R-L Series Circuits

DEFINITIONS & LAWS

- **Impedance** Z is the total opposition of an a.c. circuit to the alternating current; it is the ratio of the r.m.s. (or peak) voltage across the circuit to the r.m.s. (or peak) current in it.
- In a series R-L circuit, the voltage leads the current by a phase angle φ; in a series R-C circuit, the current leads the voltage by φ, where 0° < φ < 90°.

- **Power factor** is the ratio of the mean power dissipated to the product of V_{rms} and I_{rms}, equal to $\cos\varphi = \frac{R}{Z}$.
- The mean power is dissipated only in the resistance; the inductor and capacitor dissipate no mean power.
- The voltages across R and L (or C) are 90° out of phase, so the supply voltage is their phasor (vector) sum and not their arithmetic sum.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
R-L series impedance	$Z = \sqrt{R^2 + X_L^2}$	Z = impedance (Ω); R = resistance (Ω); X_L = inductive reactance (Ω)
R-L series phase angle	$\tan\varphi = \frac{X_L}{R}$	φ = phase angle between voltage and current (rad or °)
Inductive reactance	$X_L = 2\pi fL$	X_L = inductive reactance (Ω); f = frequency (Hz); L = inductance (H)
R-C series impedance	$Z = \sqrt{R^2 + X_C^2}$	Z = impedance (Ω); R = resistance (Ω); X_C = capacitive reactance (Ω)
R-C series phase angle	$\tan\varphi = \frac{X_C}{R}$	φ = phase angle between voltage and current (rad or °)
Capacitive reactance	$X_C = \frac{1}{2\pi fC}$	X_C = capacitive reactance (Ω); f = frequency (Hz); C = capacitance (F)
Supply voltage and current	$V_{rms} = I_{rms} Z$	V_{rms} = r.m.s. supply voltage (V); I_{rms} = r.m.s. current (A); Z = impedance (Ω)
Voltage relation (phasor addition)	$V^2 = V_R^2 + V_L^2$ (or V_C^2); $V_R = IR$, $V_L = IX_L$, $V_C = IX_C$	V = supply voltage (V); V_R, V_L, V_C = voltages across R, L, C (V); I = current (A)
Power factor	$\cos\varphi = \frac{R}{Z}$	cosφ = power factor (no unit); R = resistance (Ω); Z = impedance (Ω)
Mean power	$P = V_{rms} I_{rms} \cos\varphi = I_{rms}^2 R$	P = mean power (W); V_{rms}, I_{rms} = r.m.s. voltage (V) and current (A); R = resistance (Ω)

TIPS TO ANSWER

- Draw the impedance (or phasor) triangle: R horizontal, X vertical, Z hypotenuse, and φ between R and Z; then use Pythagoras and trigonometry.
- Calculate X_L or X_C first, then $Z = \sqrt{R^2 + X^2}$, then $I_{rms} = \frac{V_{rms}}{Z}$ and finally the required voltage or power, giving each answer with a unit.

- State the phase relation clearly: 'current leads voltage by φ' for R-C, 'current lags voltage by φ' for R-L, with $\tan\varphi = \frac{X}{R}$.
- For the variation with frequency: R-L: as f increases, X_L and Z increase, so I decreases; R-C: as f increases, X_C decreases, so I increases.

SECTION A TRAPS

Trap: Adding arithmetically: Z = R + X (30 + 40 = 70 Ω). The correct value is $Z = \sqrt{30^2 + 40^2} = 50\ \Omega$.

Trap: Using $\tan\varphi = \frac{R}{X}$ (inverted). For both circuits, $\tan\varphi = \frac{X}{R}$; check that φ = 90° when R = 0.

Trap: Taking the power factor as $\frac{Z}{R}$, which exceeds 1; the power factor is $\frac{R}{Z}$ and never exceeds 1.

Trap: Adding the component voltages to get the supply voltage; $V_R + V_L$ is greater than the supply, since $V = \sqrt{V_R^2 + V_L^2}$.

TRICKS

Trick: Recognise Pythagorean triples: (30, 40, 50), (60, 80, 100) and (6, 8, 10); Z, I and the voltage across each component then follow without a calculator.

Trick: Limiting case: when R increases with X fixed, $\varphi \to 0°$ and $\cos\varphi \to 1$ (more resistive); when $R \to 0$, $\varphi \to 90°$ (purely reactive).